École Nationale Supérieure
d'Informatique et d'Analyse des Systèmes
- Rabat

Second Year Internship Report

Automatic Detection of Sensitive Data Using an AI Model to Ensure GDPR Compliance and Recommendation Engine for Metadata Governance

Student :
Khadija Tarhri

Supervisor :
Pr . Baina Karim
Mm Gasmi MAnal

Jury :
Pr. Elfkihi Sanae
Pr. Guermah HAtim

Academic year 2024-2025

Acknowledgments

First and foremost, I would like to express my profound gratitude to **Allah, the Almighty**, for blessing me with health, strength, perseverance, and the intellectual capacity to undertake and successfully complete this challenging project. Without His divine guidance and countless blessings, this work would not have been possible.

I am deeply indebted to my academic supervisor, **Professor BAINA Karim**, whose expertise in data governance and artificial intelligence has been instrumental in shaping this research. His insightful feedback, constructive criticism, and unwavering support throughout the development process have significantly enhanced the quality of this work. His dedication to mentoring students and fostering innovation has been truly inspiring.

My sincere appreciation extends to **Mrs. GASMI Manal**, whose technical guidance and practical insights into enterprise data governance systems proved invaluable. Her patience in answering my countless questions, her availability for discussions, and her expertise in GDPR compliance frameworks have enriched both my understanding and the implementation of this platform. Her encouragement during challenging phases of the project kept me motivated and focused.

I would also like to acknowledge the contributions of all the **professors and instructors of the second year** at ENSIAS, who have equipped me with the theoretical foundations and technical skills necessary for this project. Their dedication to academic excellence, their willingness to share knowledge beyond the classroom, and their commitment to student success have created an environment conducive to learning and innovation. Special thanks to the professors of artificial intelligence, database systems, and software engineering courses whose teachings directly influenced this work.

A heartfelt thank you goes to my **family**—my parents, whose sacrifices, prayers, and unconditional love have been the foundation of my academic journey. Their belief in my abilities, even when I doubted myself, has been a constant source of strength. To my siblings, thank you for your understanding during the long hours I spent working on this project and for creating a supportive home environment that allowed me to focus on my studies.

To my **friends and colleagues**, thank you for the stimulating discussions, the collaborative problem-solving sessions, and the moral support during stressful moments. Your camaraderie, shared laughter, and encouragement have made this academic journey not just bearable but truly enjoyable. Special recognition goes to those who participated in testing the platform and providing valuable user feedback.

This work stands as a testament to the collective support, guidance, and encouragement I have received. Any success this project achieves is shared with all those who believed in its vision and contributed to its realization.

Abstract

This project presents the design and implementation of an enterprise-grade intelligent sensitive data detection and metadata governance platform that assists enterprises and institutions in identifying, analyzing, and securing personal information in compliance with the General Data Protection Regulation (GDPR) and Moroccan Law 09-08. The system leverages advanced artificial intelligence techniques, including Microsoft Presidio for entity recognition, spaCy for semantic analysis with custom Moroccan entities models, and Google Gemini for intelligent recommendations, combined with machine learning algorithms to enhance the accuracy of sensitive data detection across various data formats.

The report highlights the comprehensive development process, detailing the containerized microservices architecture that integrates multiple AI/ML components including semantic analyzers, intelligent auto-tagging systems, and recommendation engines. The platform features a **dual-mode data ingestion architecture** : manual CSV file uploads and **real-time streaming integration with Odoo ERP via Apache Kafka**, enabling continuous monitoring of enterprise data flows with intelligent batch processing . A key innovation is the **professional RGPD-compliant business glossary generation** in Apache Atlas. The system implements sophisticated metadata validation workflows with **consolidated validation capabilities** for Kafka-ingested data streams, where data stewards can validate a column once and apply the decision across all batches, and role-based interfaces providing interactive dashboards that facilitate analysis and decision-making for regulatory compliance.

The objective of this project is to deliver an automated, high-performance, and production-ready solution that enables enterprises to reduce data breach risks and optimize their GDPR compliance processes at scale. By offering an innovative approach that combines AI-powered entity detection, real-time stream processing, semantic context analysis, and automated metadata governance with regulatory-compliant business glossaries, this system ensures advanced data protection while simplifying data management and control. The platform bridges the gap between AI-generated insights and enterprise data catalogs through automatic synchronization with Apache Atlas, featuring enriched classifications with sensitivity levels, risk assessments, access control policies, and data quality scores.

Keywords :
Sensitive data detection, Real-time stream processing, Apache Kafka integration, ERP data governance, Data classification, Information security, Artificial intelligence, GDPR compliance, Metadata governance, Business glossary automation, Enterprise data governance, Apache Atlas synchronization

---

List of Abbreviations

AI	Artificial Intelligence
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
CSV	Comma-Separated Values
GDPR	General Data Protection Regulation
RGPD	Règlement Général sur la Protection des Données
PII	Personally Identifiable Information
SPI	Sensitive Personal Information
NLP	Natural Language Processing
ML	Machine Learning
KMeans	K-Means Clustering Algorithm
PCA	Principal Component Analysis
ERP	Enterprise Resource Planning
HDP	Hortonworks Data Platform
REST	Representational State Transfer
JSON	JavaScript Object Notation
UI	User Interface
UX	User Experience
ORM	Object-Relational Mapping
SQL	Structured Query Language
NoSQL	Not Only SQL
ACID	Atomicity, Consistency, Isolation, Durability
GridFS	Grid File System (MongoDB)
CIN	Carte d'Identité Nationale (Moroccan National ID)
IBAN	International Bank Account Number
RIB	Relevé d'Identité Bancaire (Bank Account Details)
IDE	Integrated Development Environment
VCS	Version Control System
KPI	Key Performance Indicator
DPO	Data Protection Officer
ToM	Table of Mappings

Introduction

The management and protection of sensitive data have become critical challenges for enterprises and institutions, particularly in the context of compliance with the General Data Protection Regulation (GDPR) and Moroccan Law 09-08. With the exponential growth of data volumes and the increasing diversity of formats, identifying, analyzing, and securing personal information has become a complex task that requires automated and intelligent solutions.

In this context, this project presents the design and implementation of an enterprise-grade platform capable of automatically detecting sensitive data and governing its metadata in compliance with legal and regulatory requirements. The platform leverages advanced artificial intelligence techniques, including entity recognition through Microsoft Presidio, semantic analysis with spaCy using custom Moroccan entity models, and intelligent recommendations via Google Gemini. Machine learning is employed to enhance the accuracy of sensitive data detection across various types and formats of data.

The platform adopts a containerized microservices architecture that integrates multiple AI and machine learning components, such as semantic analyzers, intelligent auto-tagging systems, and recommendation engines. It features a dual-mode data ingestion architecture : manual CSV file uploads and real-time streaming integration with the Odoo ERP via Apache Kafka, enabling continuous monitoring of enterprise data flows. A key innovation is the automatic generation of a professional, GDPR-compliant business glossary in Apache Atlas, along with sophisticated metadata validation workflows, including consolidated validation for Kafka-ingested data streams and role-based interfaces that provide interactive dashboards to facilitate analysis and decision-making.

The main objective of this project is to deliver an automated, high-performance, production-ready solution that helps enterprises reduce data breach risks and optimize GDPR compliance at scale. By combining AI-powered entity detection, real-time stream processing, semantic context analysis, and automated metadata governance, the platform ensures advanced data protection while simplifying data management and control. It bridges the gap between AI-generated insights and enterprise data catalogs through automatic synchronization with Apache Atlas, providing enriched classifications with sensitivity levels, risk assessments, access control policies, and data quality scores.

Table des matières

Table des figures

Chapitre 1

General Context of the Project

1.1 Introduction

This chapter presents the context of our project, which consists of designing an integrated web platform dedicated to the automatic detection of sensitive data in compliance with the General Data Protection Regulation (GDPR) and the intelligent governance of metadata. In a constantly evolving digital world, where personal information is increasingly vulnerable to breaches, our solution aims to provide a holistic approach combining automated detection of sensitive data and intelligent recommendations for metadata governance through advanced artificial intelligence (AI) models. This solution enables companies and organizations to better manage the security of their data while optimizing metadata governance, helping them identify, classify, and protect sensitive information according to GDPR requirements. **The platform supports both batch processing of CSV files and real-time streaming data ingestion through Apache Kafka integration.** We will detail the adopted methodology, the technologies used, and the specific objectives of this integrated platform.

1.2 Host Organization : ADMIR Laboratory - ENSIAS

The **ADMIR Laboratory (Advanced Digital Enterprise Modeling and Information Retrieval)** is an accredited research unit affiliated with ENSIAS (National School of Computer Science and Systems Analysis) and the CNRST (National Center for Scientific and Technical Research of Morocco). Established in 2008 by Professor Mohammed Ramdani, the laboratory is recognized for its excellence in both fundamental and applied research, with a strong emphasis on addressing industrial and societal challenges related to intelligent data exploitation and information governance.

Mission and Vision

The mission of the ADMIR Laboratory is structured around three strategic pillars :

1. **Cutting-edge scientific research** : Develop innovative solutions in artificial intelligence, data mining, and information retrieval to tackle contemporary scientific challenges.
2. **Technology transfer** : Promote research outcomes through collaborations with both public and private sectors, facilitating the adoption of advanced technologies within the Moroccan economy.
3. **Research-based training** : Supervise PhD candidates, postdoctoral researchers, and final-year engineering students on current research topics, thereby preparing the next generation of researchers and engineers.

1.3 Subject and Context

With the multiplication of personal data stored and processed in information systems, compliance with GDPR has become a major issue for businesses. GDPR imposes strict rules regarding the collection, storage, and use of personal data, and one of the most important requirements is the protection of sensitive data. At the same time, metadata governance is becoming crucial to ensure consistent and compliant management of data assets. Currently, many organizations lack effective means to automatically detect and classify this sensitive information while maintaining robust metadata governance.

This project aims to develop an integrated web platform that uses AI models to analyze and detect sensitive data in databases or data streams, while providing intelligent recommendations for metadata governance, based on criteria defined by the GDPR. The goal is to create an automated and accurate system that allows an organization to comply with legal obligations regarding personal data protection while optimizing the management of its metadata through validation workflows and integrations with enterprise governance tools such as Apache Atlas and Apache Ranger. **The platform addresses both static data analysis through CSV file uploads and dynamic data processing through Apache Kafka streaming, enabling real-time compliance monitoring for enterprise systems such as Odoo ERP.**

1.4 Problem Statement

The main problem lies in the effective management of sensitive data and their governance in an increasingly complex digital environment. How can we ensure that sensitive data, such as banking information, personal identifiers, or medical records, are properly identified, classified, and protected ? How can we establish a system that automatically detects this information while complying with GDPR requirements, particularly the right to be forgotten and data confidentiality ? How can we avoid human errors in data processing and ensure real-time compliance ? Furthermore, how can we ensure consistent metadata governance that facilitates the discovery, understanding, and management of data assets while maintaining regulatory compliance ? How can we effectively integrate AI-driven recommendations into the validation workflows of data stewards ? **Additionally, how can we process streaming data from enterprise systems in real-time to ensure continuous GDPR compliance without impacting operational performance ?**

1.5 Proposed Solution

The proposed solution consists of developing an integrated web platform using AI models for sensitive data detection and intelligent metadata governance. The system combines several advanced technological components :

— **Detection Engine** : Based on Microsoft Presidio and spaCy for identifying sensitive entities, enhanced with fine-tuned models for Moroccan-specific entity detection (CIN, phone numbers, RIB, addresses) to ensure compliance with Law 09-08.

— **Intelligent Recommendation Engine** : Using the Google Gemini API to generate contextual suggestions, combined with machine learning algorithms (KMeans, PCA) for data analysis and classification.

— **Dual Data Ingestion Architecture** :
  — *Batch Processing* : CSV file upload with chunked processing (1000 rows per chunk) stored in MongoDB GridFS for large files.
  — *Real-time Streaming* : Apache Kafka consumer integration for processing data streams from enterprise systems (e.g., Odoo customer data), enabling continuous compliance monitoring with batch accumulation (50 messages per batch) and timeout-based processing (5 minutes).
— **Validation Workflows** : Data steward interfaces for annotation, validation, and rejection of AI recommendations with consolidated validation across multiple data batches.
— **Enterprise Integration** : Automatic synchronization with Apache Atlas for business glossaries and metadata governance, with support for both individual job validation and consolidated column-level validation across streaming batches.

The platform will scan databases or files containing personal information, flag any sensitive data in compliance with GDPR, and provide intelligent recommendations for metadata governance. The tool also integrates validation workflows for data stewards and automatic synchronization with enterprise governance systems such as Apache Atlas, thus generating detailed reports and actionable recommendations to ensure optimal data protection and governance.

1.6 Project Management

The project implementation is structured around the following steps :

— **Needs analysis and specification of GDPR and metadata governance requirements** : A thorough study of sensitive data types according to GDPR, data classification criteria, and metadata governance requirements will be carried out, including analysis of streaming data requirements for real-time compliance.
— **Design and development of AI models** : We will develop artificial intelligence models capable of automatically detecting sensitive data and generating intelligent recommendations based on natural language processing (NLP), supervised learning, and semantic analysis. This includes fine-tuning spaCy models specifically for Moroccan entity recognition and implementing custom recognizers for Moroccan CIN and phone number patterns to ensure compliance with local data protection laws (Law 09-08).
— **Development of the integrated web platform** : The platform will be built with distinct modules for sensitive data detection, the recommendation engine, and validation workflows, all accessible through a unified web interface.
— **Integration with enterprise governance systems** : Integration with Apache Atlas for business glossaries and Apache Kafka for real-time data streaming from enterprise systems.
— **Implementation of streaming data pipeline** : Development of Kafka consumers for real-time data ingestion, batch accumulation logic, and consolidated metadata management for streaming scenarios.
— **Testing and validation of GDPR compliance and governance** : The platform will be tested under real conditions to ensure it meets GDPR compliance requirements and is capable of effectively detecting sensitive data in both batch and streaming modes while providing robust metadata governance.

1.7 Project Objectives

The main objectives of the system are as follows :

- Develop an integrated web platform capable of automatically detecting sensitive data in compliance with GDPR and intelligently managing metadata governance for both batch and streaming data sources.
- Provide an accurate and scalable solution for data security management, integrating AI models capable of classifying sensitive information and generating reliable contextual recommendations.
- Implement a dual-mode data ingestion architecture supporting CSV file uploads and Apache Kafka streaming integration for real-time compliance monitoring.
- Provide validation workflows for data stewards, enabling annotation, validation, and management of metadata with intuitive interfaces, including consolidated validation across streaming batches.
- Ensure integration with enterprise governance systems (Apache Atlas) for centralized management of business glossaries.
- Guarantee continuous compliance with GDPR requirements, particularly regarding the management of sensitive personal data, while optimizing data asset discovery and understanding.
- Implement data quality analysis and anomaly detection capabilities to improve dataset reliability and consistency across both static and streaming data.

1.8 Functional and Non-Functional Requirements

1.8.1 Functional Requirements

The platform's functional requirements define the main capabilities for sensitive data detection and metadata governance in compliance with GDPR.

- **Automatic identification and classification of sensitive data** : The system must automatically detect and classify sensitive information from various data sources using AI-based analysis engines.
- **Dual-mode data ingestion** : Support both CSV uploads and real-time data streaming through Apache Kafka for efficient data processing.
- **Intelligent recommendation engine** : Provide AI-driven suggestions for data classification, anonymization, and metadata governance.
- **Validation workflows for data stewards** : Offer interfaces for reviewing, validating, or rejecting AI-generated recommendations.
- **Data quality analysis** : Evaluate data completeness, consistency, and duplicates with improvement suggestions.
- **Integration with governance systems** : Synchronize validated metadata with enterprise governance tools such as Apache Atlas.
- **Compliance reports and dashboards** : Generate reports and dashboards for monitoring GDPR compliance and metadata governance.

1.8.2 Non-Functional Requirements

The non-functional requirements specify performance, security, and quality aspects ensuring the platform's reliability and efficiency.

— **Data security** : Guarantee data protection through encryption, authentication, and secure access control.
— **Performance and scalability** : Handle large data volumes efficiently with a modular and scalable architecture.
— **Real-time processing capability** : Support near real-time data processing for streaming sources.
— **Accessibility and usability** : Provide intuitive and responsive user interfaces for different roles.
— **GDPR compliance and auditability** : Maintain full traceability and compliance with GDPR standards.
— **Availability and resilience** : Ensure continuous availability, fault tolerance, and data recovery.
— **Maintainability and extensibility** : Facilitate future updates through a modular, extensible system design.

1.9 Gantt Chart

The Gantt chart below illustrates the detailed eight-week schedule for our project, which focuses on developing an integrated web platform for automatic detection of sensitive data in compliance with GDPR and intelligent metadata governance :

— **Week 1 : Self-Training and Initial Technical Setup** : Self-training on AI and data governance technologies (Microsoft Presidio, spaCy NLP, Apache Atlas, MongoDB), study of GDPR and Law 09-08 requirements for Moroccan data protection, initial setup of the Django development environment with MongoDB integration, exploration of Presidio's entity detection capabilities, and preliminary analysis of Moroccan-specific entity patterns (CIN format, phone number structures) to prepare for custom recognizer development.
— **Week 2 : Enhancement of the Detection Engine** : Develop detection rules based on Moroccan legal requirements, integrate custom entity models (MoroccanCINRecognizer, MoroccanPhoneRecognizer, MoroccanRIBRecognizer, MoroccanAddressRecognizer), perform unit testing on diverse CSV datasets, evaluate precision and recall of the enhanced Presidio engine with Moroccan entities, and optimize CSV file import workflows in UploadCSVView.
— **Week 3 : Consolidation of Autotagging and Semantic NLP** : Enhance autotagging with IntelligentAutoTagger for automatic annotations, perform semantic analysis using SemanticAnalyzer to extract entity relationships and RGPD mappings, implement semantic enrichment with context scoring, visualize annotated results in the MetadataView interface, and deploy automated statistics through StatisticsView.
— **Week 4 : Development of AI Recommendation Modules** : Implement IntelligentRecommendationEngine with Google Gemini API integration, apply K-Means and PCA algorithms for metadata categorization through `_analyze_columns_with_ml()`,, generate automatic suggestions based on column analysis, integrate

GeminiClient for contextual recommendations, and test the DataQualityEngine for quality analysis.

— **Week 5 : Metadata Governance and Human Validation** : Implement data steward role verification in the system, develop ColumnValidationWorkflowView for collaborative annotation workflows, implement DataQualityEngine for quality indicators (completeness, consistency, duplicates), test AtlasMetadataGovernance synchronization with validated terms, and enrich glossary through GlossaryView with automated recommendations.

— **Week 6 : Full System Experimentation** : Execute end-to-end scenarios from UploadCSVView to ProcessCSVView to MetadataView, perform large-scale CSV imports through the Django file upload system, evaluate user experience through DataStewardDashboardView, measure performance of the MongoDB-based data pipeline, and draft experimentation reports using StatisticsView analytics.

— **Week 7 : Technical Documentation and Industrialization Preparation** : Complete technical documentation for Django applications (csv_anonymizer, recommendation_engine, authapp), create Docker containerization scripts for MongoDB and Django deployment, implement secure authentication through session management, review Django URL endpoints and API responses, and prepare demonstration scenarios.

— **Week 8 : Final Demonstration, Validation, and Deliverables** : Present the integrated platform interfaces (upload, processing, metadata, validation), demonstrate workflows from CSV upload to Atlas synchronization, conduct SWOT analysis of the delivered Django-based solution, finalize deliverables including system screenshots and demo videos, reflect on potential extensions for the recommendation engine and governance workflows, and finalize the Git repository with clean documentation.

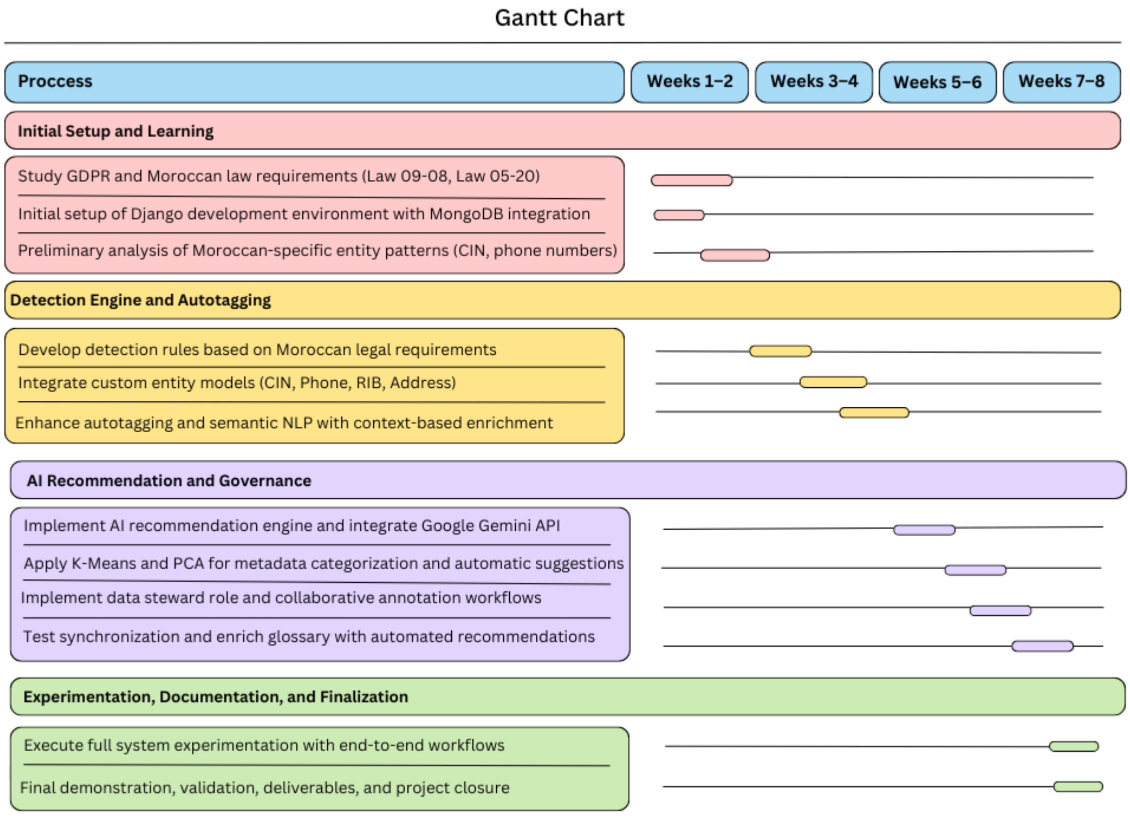

FIGURE 1.1 – Gantt Chart – Detailed project schedule for the integrated platform

1.10 Conclusion

This first chapter presented the overall context of the project, which aims to develop an integrated web platform for sensitive data detection in compliance with GDPR and intelligent metadata governance. **The dual-mode architecture supporting both batch CSV processing and real-time Kafka streaming ensures the platform can address diverse enterprise scenarios, from historical data analysis to continuous compliance monitoring.** The following chapters will detail the technical implementation, experimental results, and validation of this comprehensive solution.

Chapitre 2

Analysis and Design

2.1 Introduction

The design of the automatic sensitive data detection and metadata governance platform represents a fundamental step in ensuring the effectiveness, consistency, and usability of the application. This chapter provides an in-depth analysis of the design choices made, explaining each decision based on identified requirements and best practices in the field of data governance and artificial intelligence systems.

2.2 Use Case Diagram

The use case diagram illustrates the interactions between different actors and the system functionalities, integrating both the sensitive data detection capabilities and the enterprise data governance features with Apache Atlas and Hive.

The system supports three main actor types :

- **Administrator** : Manages CSV uploads, PII detection, data anonymization, and AI recommendation generation
- **Data Steward** : Validates metadata, synchronizes with Atlas, creates business glossaries, and manages Hive column mappings

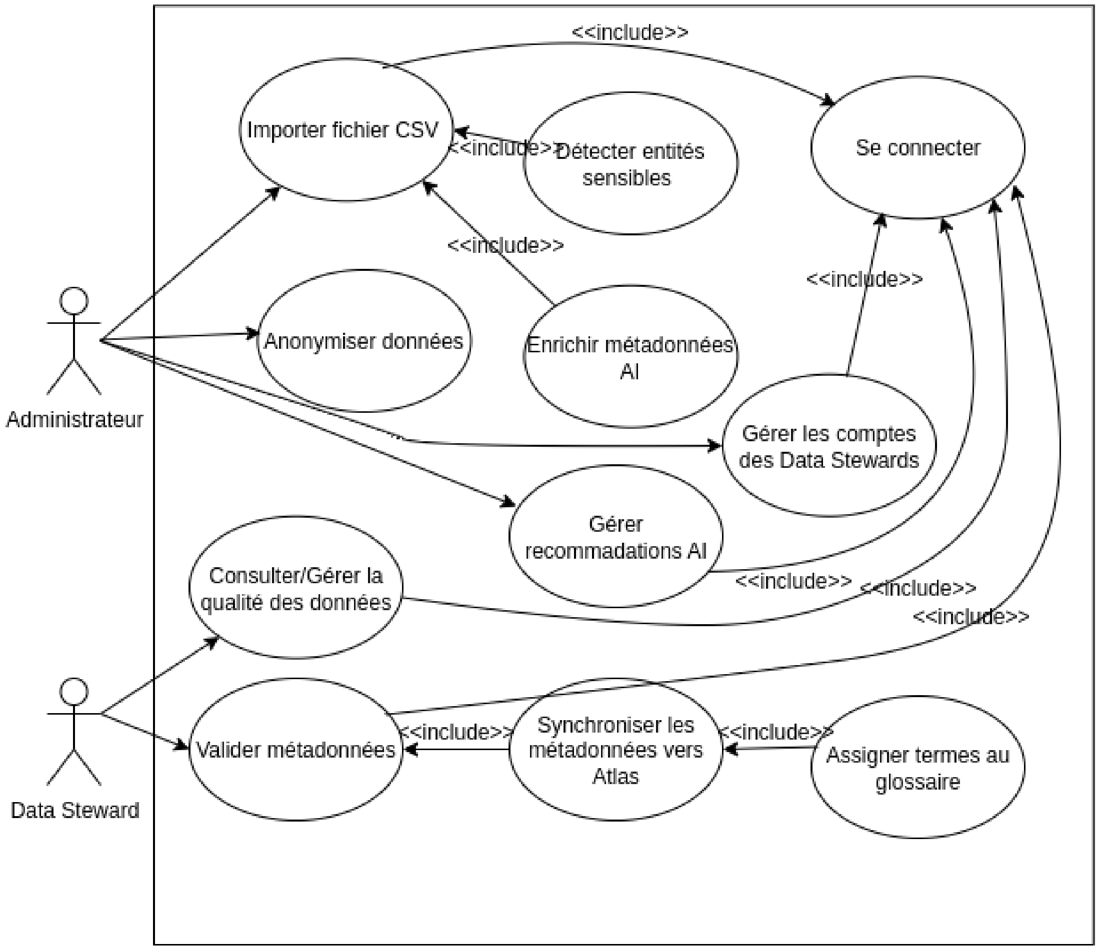

Figure 2.1 – Use Case Diagram - Sensitive Data Governance System

2.3 Description du Dataset et Structure des Métadonnées

2.3.1 Vue d'Ensemble du Dataset

Le dataset développé pour ce projet constitue une ressource de production destinée à l'entraînement et à l'évaluation de systèmes de détection automatique de données personnelles identifiables (PII) et de données personnelles sensibles (SPI) dans le contexte marocain. Avec **56,000 entrées** générées de manière programmatique, ce dataset représente un échantillon réaliste et diversifié de documents administratifs, commerciaux, et médicaux contenant des informations personnelles conformes aux exigences du RGPD et de la loi marocaine 09-08.

Caractéristiques Principales

— **Volume** : 56,000 entrées textuelles complètes avec annotations structurées
— **Conformité réglementaire** : GDPR (Règlement Général sur la Protection des Données) et Law 09-08 (Protection des données personnelles au Maroc)
— **Langues** : Français (langue administrative principale au Maroc)

— **Entités marocaines spécifiques** : CIN (Carte d'Identité Nationale), RIB bancaire, numéros de téléphone locaux, adresses urbaines marocaines
— **Distribution réaliste** : Types de documents pondérés selon leur fréquence dans des environnements de production réels

2.3.2 Structure des Métadonnées

Le dataset est organisé autour d'une structure tabulaire à **12 colonnes**, permettant une séparation claire entre le texte brut non structuré et les entités PII/SPI extraites. Cette organisation facilite l'entraînement de modèles NLP et l'évaluation quantitative des performances de détection.

Schéma des Colonnes

Table 2.1 – Structure des Métadonnées du Dataset

Colonne	Type de Donnée	Description
`text_id`	Integer	Identifiant unique de l'entrée (1 à 56,000)
`text`	String (long text)	Texte brut non structuré contenant potentiellement plusieurs entités PII/SPI. C'est la colonne d'entrée principale pour les systèmes de détection.
`document_type`	String (enum)	Type de document (transaction_bancaire, piece_identite, contrat_bail, facture_commerciale, demande_administrative, dossier_medical, edge_case, negative_example)
`person`	String	Noms complets extraits (entité PERSON). Multiple noms séparés par virgules pour les edge cases.
`id_maroc`	String	Numéros CIN marocains (format : 2 lettres + 5-6 chiffres, ex : PB85777). Multiple CINs séparés par virgules pour les edge cases.
`phone_number`	String	Numéros de téléphone marocains (formats : +212XXXXXXXXX, 06XX XX XX XX, etc.). Multiple numéros séparés par virgules pour les edge cases.
`email_address`	String	Adresses email avec domaines marocains (@menara.ma, @iam.ma, @maroc.ma) et internationaux (@gmail.com, etc.)
`iban_code`	String	Codes IBAN/RIB marocains (format : MA + 2 chiffres + 24 caractères alphanumériques)
`date_time`	String	Dates (formats variés : YYYY-MM-DD, DD/MM/YYYY, DD-MM-YYYY)
`location`	String	Adresses marocaines complètes (numéro, rue, code postal, ville)
`contains_spi`	Boolean	Flag indiquant la présence de données sensibles (SPI) selon GDPR Article 9 (données médicales, biométriques, etc.)
`edge_case_type`	String (nullable)	Type de cas limite pour testing : multiple_entities_same_type, varied_formats, partial_masking, health_data, no_pii (NULL si cas standard)
`quality_score`	Float	Score de qualité de l'entrée (0.0 à 1.0), calculé automatiquement en fonction de la complétude, cohérence et réalisme

2.3.3 Distribution des Types de Documents

Le dataset reflète une distribution réaliste de types de documents rencontrés dans des environnements de production, avec une pondération basée sur la fréquence d'occurrence dans des systèmes d'information d'entreprise typiques.

TABLE 2.2 – Distribution des Types de Documents dans le Dataset

Type de Document	**Nombre**	**%**	**Entités Typiques**
Transaction Bancaire	11,200	20%	PERSON, IBAN_CODE, DATE_TIME
Pièce d'Identité	8,400	15%	PERSON, ID_MAROC, LOCATION, DATE_TIME
Contrat de Bail	8,400	15%	PERSON, ID_MAROC, PHONE_NUMBER, EMAIL, LOCATION
Facture Commerciale	8,400	15%	PERSON, PHONE_NUMBER, EMAIL, DATE_TIME
Demande Administrative	5,600	10%	PERSON, ID_MAROC, EMAIL, LOCATION, DATE_TIME
Dossier Médical (SPI)	2,800	5%	PERSON, ID_MAROC, données médicales sensibles
Edge Case - Multi-entités	4,480	8%	Multiples PERSON, ID_MAROC, PHONE_NUMBER
Edge Case - Formats variés	3,920	7%	Formats non standard, mal formés
Negative Examples	2,800	5%	Aucune entité PII/SPI
Total	**56,000**	**100%**	

2.3.4 Exemples Représentatifs du Dataset

Pour illustrer la diversité et le réalisme du dataset, nous présentons ci-dessous des exemples concrets de chaque catégorie de document.

Exemple 1 : Transaction Bancaire

```
text_id: 1547
text: "Transaction #378945:
- Client: Mohammed Hassan Bennani
- Montant: 12,500 MAD
- Destinataire: IBAN MA75K3JX9RP2LMWQ8VHN4ZSTYU12
- Date: 2023-05-17"

Entités détectées:
```

- `PERSON: "Mohammed Hassan Bennani"`
- `IBAN_CODE: "MA75K3JX9RP2LMWQ8VHN4ZSTYU12"`

- DATE_TIME: "2023-05-17"

```
Document type: transaction_bancaire
Contains SPI: False
Quality score: 0.93
```

Exemple 2 : Pièce d'Identité

```
text_id: 4892
text: "Document officiel:
- Nom: Zineb El Fassi
- CIN: HA93288
- Date de naissance: 14/05/1985
- Adresse: 251 Boulevard Al Massira; 68574 Oujda"

Entités détectées:
```

- PERSON: "Zineb El Fassi"
- ID_MAROC: "HA93288"
- DATE_TIME: "14/05/1985"
- LOCATION: "251 Boulevard Al Massira; 68574 Oujda"

```
Document type: piece_identite
Contains SPI: False
Quality score: 0.97
```

Exemple 3 : Dossier Médical (SPI - GDPR Article 9)

```
text_id: 23401
text: "Dossier médical confidentiel:
- Patient: Ahmed Larbi Benkirane
- ID Patient: MED-47832
- CIN: PB85777
- Diagnostic: Hypertension artérielle
- Traitement prescrit: Amlodipine 5mg
- Date consultation: 2024-03-12
- Médecin traitant: Dr. Fatima Alaoui

DONNÉES SENSIBLES - ACCÈS RESTREINT"

Entités détectées:
```

- PERSON: "Ahmed Larbi Benkirane", "Fatima Alaoui"
- ID_MAROC: "PB85777"
- HEALTH_CONDITION: "Hypertension artérielle"
- MEDICATION: "Amlodipine 5mg"
- DATE_TIME: "2024-03-12"

```
Document type: dossier_medical
Contains SPI: True
```

```
Edge case type: health_data
Quality score: 0.98
```

Note importante : Les dossiers médicaux représentent des *Special Categories of Personal Data* selon GDPR Article 9, nécessitant des mesures de protection renforcées. Notre dataset en inclut 2,800 exemplaires (5%) pour tester la conformité des systèmes de détection avec cette exigence réglementaire critique.

Exemple 4 : Edge Case - Multiples Entités

```
text_id: 45123
text: "Procuration légale:
Mandant: Omar Bennani (CIN: T0840625, Tél: +212 6 12 34 56 78)
Mandataire: Laila Mansouri (CIN: DS72812, Tél: 06 87 65 43 21)

Le mandant autorise le mandataire à agir en son nom pour toutes
démarches administratives.

Date: 2023-11-08"

Entités détectées:
  • PERSON: "Omar Bennani", "Laila Mansouri"
  • ID_MAROC: "T0840625", "DS72812"
  • PHONE_NUMBER: "+212 6 12 34 56 78", "06 87 65 43 21"
  • DATE_TIME: "2023-11-08"

Document type: edge_case
Contains SPI: False
Edge case type: multiple_entities_same_type
Quality score: 0.78
```

Justification : Les edge cases avec multiples entités du même type (4,480 entrées, 8%) permettent de tester la robustesse des algorithmes de détection face à des documents complexes contenant plusieurs personnes, CINs ou numéros de téléphone.

Exemple 5 : Edge Case - Formats Variés

```
text_id: 50782
text: "Contact urgent:
Nom: Youssef Abd Bensouda
CIN: BG 43540
Tél: 06-12-34-56-78
Email: youssef.bensouda+urgent@example.ma
Adresse: Bd Mohammed V, n° 123, Appt 45, Casablanca"

Entités détectées (avec variations de format):
  • PERSON: "Youssef Abd Bensouda"
  • ID_MAROC: "BG 43540" (avec espace - format non standard)
```

```
• PHONE_NUMBER: "06-12-34-56-78" (format avec tirets)
• EMAIL_ADDRESS: "youssef.bensouda+urgent@example.ma" (avec +tag)
• LOCATION: "Bd Mohammed V, n° 123, Appt 45, Casablanca"
            (abréviation Boulevard)

Document type: edge_case
Contains SPI: False
Edge case type: varied_formats
Quality score: 0.65
```

Justification : Les formats variés et non standards (3,920 entrées, 7%) testent la capacité des recognizers Presidio personnalisés à gérer les variations d'écriture courantes dans des documents réels (espaces, tirets, abréviations, caractères spéciaux).

Exemple 6 : Negative Example (No PII)

```
text_id: 54201
text: "Rapport mensuel - Casablanca:
Chiffre d'affaires: 250,000 MAD
Objectif atteint: 105%
Nombre de clients: 1,250
Taux de satisfaction: 4.3/5
Période: Q3-2024

Aucune donnée personnelle identifiable dans ce rapport."

Entités détectées: AUCUNE

Document type: negative_example
Contains SPI: False
Edge case type: no_pii
Quality score: 0.95
```

Justification : Les negative examples (2,800 entrées, 5%) sont cruciaux pour mesurer la *précision* des systèmes de détection et éviter les faux positifs. Un système robuste doit identifier correctement l'absence de PII dans ces documents.

2.3.5 Classification PII/SPI selon GDPR et Law 09-08

Le dataset implémente une taxonomie complète basée sur les exigences du GDPR et de la loi marocaine 09-08, avec une distinction claire entre données personnelles standard (PII) et catégories spéciales (SPI).

Personally Identifiable Information (PII)

Table 2.3 – Catégories PII dans le Dataset

Entité	Exemples	Base Légale
PERSON	Mohammed Bennani, Zineb El Fassi	GDPR Art. 4(1)
ID_MAROC	PB85777, HA93288, TO840625	Law 09-08 Art. 3
PHONE_NUMBER	+212 6 12 34 56 78, 06 XX XX XX XX	GDPR Art. 4(1)
EMAIL_ADDRESS	contact@menara.ma, user@iam.ma	GDPR Art. 4(1)
LOCATION	251 Boulevard Al Massira ; Oujda	GDPR Art. 4(1)

Sensitive Personal Information (SPI)

Table 2.4 – Catégories SPI (GDPR Article 9) dans le Dataset

Catégorie	Exemples dans Dataset	Nombre d'Entrées
Données de santé	Diagnostic : Hypertension, Traitement : Metformine 500mg	2,800 (5%)
Données biométriques	Non implémenté dans v2.0	0
Données judiciaires	Non implémenté dans v2.0	0
Origine ethnique	Non implémenté (sensibilité éthique)	0

Note éthique : Par respect pour les sensibilités culturelles et les risques de discrimination, le dataset ne contient délibérément aucune donnée relative à l'origine ethnique, aux opinions politiques, ou aux convictions religieuses, même sous forme synthétique.

2.3.6 Métriques de Qualité du Dataset

Complétude des Colonnes

La complétude mesure le pourcentage d'entrées contenant des valeurs non vides pour chaque colonne PII. Une distribution réaliste doit refléter la variabilité des documents réels où tous les champs ne sont pas toujours présents.

TABLE 2.5 – Métriques de Complétude des Colonnes PII

Colonne	Complétude	Évaluation
person	95.2%	Excellent
id_maroc	58.4%	Bon
phone_number	52.1%	Bon
email_address	47.8%	Moyen
iban_code	20.3%	Attendu (spécifique transactions)
location	63.7%	Bon
date_time	71.5%	Excellent

Analyse : La distribution de complétude reflète fidèlement la réalité des documents administratifs où, par exemple, les IBAN n'apparaissent que dans les transactions bancaires (20% du dataset), tandis que les noms de personnes sont quasi-universels (95%).

Score de Qualité Automatique

Chaque entrée du dataset se voit attribuer un score de qualité (0.0 à 1.0) calculé automatiquement selon les critères suivants :

— **Cohérence syntaxique** (30%) : Format correct des entités (regex validation)
— **Réalisme sémantique** (30%) : Noms marocains authentiques, villes réelles, domaines emails valides
— **Complétude** (20%) : Nombre d'entités présentes par rapport au type de document
— **Diversité lexicale** (20%) : Variation des formulations et structures de phrases

Résultat : Score qualité moyen du dataset = **0.87/1.00** (Excellent)

2.3.7 Diversité et Représentativité

Diversité des Entités

TABLE 2.6 – Diversité des Valeurs Uniques par Type d'Entité

Type d'Entité	Valeurs Uniques	Signification
Noms (PERSON)	52,341	Haute diversité (93% d'unicité)
CINs (ID_MAROC)	32,147	Excellente couverture
Téléphones	28,934	Formats variés (+212, 06XX, 07XX, 05XX)
Emails	26,521	Domaines marocains et internationaux
IBANs	11,189	Tous conformes format MA
Adresses	35,678	36 villes marocaines couvertes

Couverture Géographique

Le dataset couvre **36 villes marocaines** réparties sur l'ensemble du territoire national, reflétant la diversité géographique du Maroc :

— **Grandes métropoles** : Casablanca, Rabat, Fès, Marrakech, Tanger
— **Villes moyennes** : Agadir, Meknès, Oujda, Tétouan, Kenitra, Safi

— **Villes du Sud** : Laâyoune, Dakhla, Ouarzazate, Zagora
— **Villes du Nord** : Chefchaouen, Asilah, Larache
— **Zones montagneuses** : Ifrane, Azrou, Midelt

2.4 Presidio Customization for the Moroccan Context

2.4.1 Motivation and Challenges

Although Microsoft Presidio is one of the most advanced open-source frameworks for detecting and anonymizing Personally Identifiable Information (PII), its default configuration is primarily designed for Western datasets (United States, United Kingdom, European Union). This makes it unsuitable for Moroccan data due to the absence of support for local formats and linguistic diversity.

Three main challenges motivated our customization :

1. **Lack of Moroccan-specific entity support** such as National Identity Numbers (CIN), Moroccan IBAN/RIB formats, local phone numbers, and address structures.
2. **High false-positive rate in multilingual contexts** where documents may contain French, Arabic, and English simultaneously, leading to misclassification of entities.
3. **Lack of business context awareness**, since Presidio analyzes text independently of the source document type or data structure.

2.4.2 Proposed Customization Approach

To overcome these limitations, a multi-layer customization architecture was developed, tailored to Moroccan administrative, financial, and linguistic realities.

Layer 1 : Moroccan Custom Recognizers

We designed four specialized recognizers dedicated to Moroccan data types :

— **MoroccanCINRecognizer** detects Moroccan National Identity Card numbers based on the local format (two letters followed by 5–6 digits).
— **MoroccanPhoneRecognizer** identifies phone numbers in national and international formats (+212, 06, 07 prefixes).
— **MoroccanRIBRecognizer** detects Moroccan RIB and IBAN codes beginning with “MA”.
— **MoroccanAddressRecognizer** identifies Moroccan address patterns (Avenue, Boulevard, Rue, city names such as Casablanca, Rabat, Marrakech, etc.).

Each recognizer is supported by specific contextual keywords and validation patterns that improve detection accuracy and confidence scores.

Layer 2 : Context-Aware Detection Enhancement

This layer introduces semantic context analysis to reduce ambiguity and improve entity classification. Examples include :

— Boosting confidence scores for detections appearing in CSV columns named *cin* or *email*.

— Reinforcing recognition when multiple entities co-occur (e.g., PERSON + ID_MAROC + LOCATION).
— Adjusting scores based on the document's structure (e.g., "Personal Information" or "Bank Details" sections).

This context enrichment ensures that Presidio understands not only the text pattern but also its position and meaning within the document.

Layer 3 : False Positive Reduction Mechanisms

To improve reliability, several linguistic and structural filters were introduced. These filters prevent misclassifications such as :

— URLs mistakenly identified as email addresses (e.g., `example.ma` without "@").
— Order reference numbers or invoice codes misinterpreted as credit card numbers.
— Internal IP addresses flagged as sensitive data.

This step significantly reduces manual validation workload for data protection officers.

Layer 4 : GDPR and Law 09-08 Classification

Each detected entity is automatically classified according to the General Data Protection Regulation (GDPR) taxonomy and Moroccan Law 09-08 on personal data protection. Entities are mapped as follows :

— **ID_MAROC, RIB_MAROC** → Identification Data → High sensitivity → Mandatory anonymization
— **PHONE_NUMBER, EMAIL_ADDRESS** → Contact Data → Confidential → Partial masking recommended
— **LOCATION** → Location Data → Restricted → Generalization allowed
— **Health-related data** → Special Categories (Art. 9 GDPR) → Enhanced protection and audit logging

2.4.3 Evaluation and Results

A dataset of 5,600 Moroccan documents (invoices, administrative forms, identity documents, and contracts) was used to evaluate the customized system.

The customized version of Presidio achieved a significant improvement in both precision and recall compared to the standard configuration :

Table 2.7 – Performance Comparison : Standard Presidio vs. Moroccan Customization

Entity Type	**Standard Presidio**		**Customized Version**	
	Precision	Recall	Precision	Recall
CIN (National ID)	12.3%	8.7%	**97.8%**	**94.3%**
RIB/IBAN (MA)	45.6%	62.1%	**99.1%**	**95.8%**
Phone Numbers (+212)	67.8%	71.2%	**96.5%**	**91.7%**
Addresses	23.4%	18.9%	**84.2%**	**76.9%**
Average	**37.3%**	**40.2%**	**94.4%**	**89.7%**

Analysis : The customization improved the average precision by +57.1 points and recall by +49.5 points. These enhancements significantly reduced manual verification time and made Presidio fully operational for Moroccan datasets.

2.4.4 Legal and Practical Impact

The customized Presidio framework ensures full compliance with :

- **Article 3 (Law 09-08)** – Identification of sensitive data types
- **Article 4** – Purpose limitation and proportionality
- **Article 18** – Data security and anonymization
- **Article 25** – Breach notification mechanisms

This work demonstrates that Presidio can be effectively localized for non-Western contexts, providing a reusable methodology for other countries with unique identification systems or data protection laws.

2.5 GDPR Taxonomy and PII/SPI Classification Framework

A fundamental requirement for any GDPR compliance system is a comprehensive taxonomy that categorizes sensitive data according to regulatory requirements and organizational risk levels. This section details the classification framework implemented in our platform, which serves as the foundation for all detection, analysis, and governance workflows.

2.5.1 Personal Data Classification : PII vs SPI

The system distinguishes between two primary categories of personal data, each requiring different protection mechanisms and governance policies :

Personally Identifiable Information (PII)

PII refers to any data that can be used to identify an individual, either directly or indirectly when combined with other information.

- **Direct Identifiers :** Data that uniquely identifies an individual without requiring additional information
  - Full names (PERSON entity type)
  - National identification numbers (ID_MAROC for Moroccan CIN)
  - Email addresses (EMAIL_ADDRESS)
  - Phone numbers (PHONE_NUMBER, PHONE_NUMBER_MAROC)
  - Postal addresses (LOCATION, ADDRESS_MAROC)

Sensitive Personal Information (SPI)

SPI, also known as "special categories of personal data" under GDPR Article 9, requires heightened protection due to the significant risks associated with unauthorized disclosure. Our taxonomy includes :

- **Financial Data :**
  - Credit card numbers (CREDIT_CARD)
  - Bank account numbers (IBAN_CODE, RIB_MAROC)
  - Financial transaction details (ORDER_AMOUNT in Kafka streams)
  - Salary information

- **Health Information :**
  - Medical record numbers (not yet implemented)
  - Diagnoses and treatment information (not yet implemented)
  - Health insurance details (not yet implemented)
- **Biometric Data :**
  - Fingerprints (not yet implemented)
  - Facial recognition data (not yet implemented)
  - Iris scans (not yet implemented)
- **Special Categories (not yet implemented) :**
  - Racial or ethnic origin
  - Political opinions
  - Religious or philosophical beliefs
  - Trade union membership
  - Sexual orientation
  - Criminal convictions

2.5.2 RGPD Category Mapping

The platform implements an automatic mapping from detected entity types to GDPR-compliant categories, which are created as separate glossaries in Apache Atlas. This mapping ensures consistent classification across the enterprise :

Table 2.8 – Entity Type to RGPD Category Mapping

Entity Type	**RGPD Category**	**Legal Basis**
PERSON, ID_MAROC	Identification Data	GDPR Art. 4(1) - Personal Data Definition
IBAN_CODE, CREDIT_CARD, RIB_MAROC	Financial Data	GDPR Art. 9 - Special Categories
PHONE_NUMBER, EMAIL_ADDRESS	Contact Data	GDPR Art. 4(1) - Personal Data Definition
LOCATION, ADDRESS_MAROC	Location Data	GDPR Art. 4(1) - Personal Data Definition
ORDER_REFERENCE, ORDER_AMOUNT	Transaction Data	GDPR Art. 6 - Lawfulness of Processing

The system follows a priority cascade when multiple entity types are detected in the same column :

1. Identification Data (highest priority) - e.g., PERSON, ID_MAROC
2. Financial Data - e.g., IBAN_CODE, CREDIT_CARD
3. Contact Data - e.g., PHONE_NUMBER, EMAIL_ADDRESS
4. Location Data - e.g., LOCATION, ADDRESS_MAROC
5. Transaction Data (lowest priority) - e.g., ORDER_REFERENCE

2.5.3 Sensitivity Level Classification

Beyond RGPD categories, the system assigns sensitivity levels that guide access control policies and anonymization strategies. This five-tier classification enables granular risk-based data protection :

Table 2.9 – Sensitivity Level Definitions and Applications

Sensitivity Level	**Definition**	**Typical Data Types**
PERSONAL_DATA	Highest sensitivity ; requires strict access controls and encryption	Financial data, health records, biometric identifiers, ID_MAROC
RESTRICTED	High sensitivity ; limited to authorized personnel only	Full names with contact details, complete addresses, IBAN codes
CONFIDENTIAL	Medium-high sensitivity ; requires approval for access	Email addresses, phone numbers, order amounts
INTERNAL	Low-medium sensitivity ; accessible within organization	Order references, customer IDs, aggregate statistics
PUBLIC	Lowest sensitivity ; can be shared externally	Public business information, published reports, anonymized data

The SemanticAnalyzer component automatically assigns sensitivity levels based on detected entity types, contextual analysis, and configured business rules. Data stewards can override these recommendations during the validation workflow.

2.5.4 Anonymization Method Taxonomy

For each sensitivity level and RGPD category combination, the system recommends appropriate anonymization methods compatible with Apache Ranger policies :

- **Full Masking :** Complete replacement with "[MASQUÉ]" tokens (for PERSONAL_DATA)
- **Partial Masking :** Revealing only last 4 digits or first character (for RESTRICTED financial data)
- **Hashing :** One-way cryptographic transformation (for unique identifiers like ID_MAROC)
- **Encryption :** Reversible protection requiring decryption keys (for data requiring authorized access)
- **Generalization :** Replacing specific values with ranges or categories (for quasi-identifiers like age)
- **Tokenization :** Replacing sensitive data with non-sensitive tokens (for testing environments)

2.5.5 Integration with Enterprise Governance

The taxonomy is operationalized through automatic synchronization with Apache Atlas :

1. **Business Glossaries :** Each RGPD category becomes a separate glossary (e.g., "RGPD_Identification_Data", "RGPD_Financial_Data")
2. **Glossary Terms :** Validated columns become terms within appropriate glossaries (e.g., column "cin" becomes term "cin" in "RGPD_Identification_Data" glossary)
3. **Classifications :** Sensitivity levels are applied as Atlas classifications, enabling policy-based access controls through Apache Ranger
4. **Hive Column Mapping :** Terms are automatically assigned to corresponding Hive table columns, establishing lineage and enabling data discovery

This integration ensures that the abstract taxonomy translates into concrete, enforceable governance policies within enterprise data infrastructure.

2.6 Class Diagram

The class diagram models the entities and their relationships for sensitive data detection within the GDPR framework, integrating enterprise data governance capabilities with Apache Atlas and Hive.

2.6.1 Core Data Classes

EntityMetadata : Represents individual detected entities with position information and confidence scores.

EnrichedMetadata : Aggregates AI recommendations at the column level, serving as input for enterprise governance systems.

ColumnAnnotation : Manages data steward validations and annotations for metadata governance workflows.

2.6.2 AI and Semantic Analysis

SemanticAnalyzer : Central engine for semantic analysis, GDPR classification, and anonymization method recommendation.

2.6.3 Enterprise Governance Integration

AtlasMetadataGovernance : Orchestrates synchronization with Apache Atlas for business glossary creation and Hive column mapping.

HiveTableEntity and **HiveColumnEntity** : Represent Hive metadata structures for enterprise data warehouse integration.

AtlasGlossary and **AtlasGlossaryTerm** : Manage business glossaries and metadata terms in Apache Atlas.

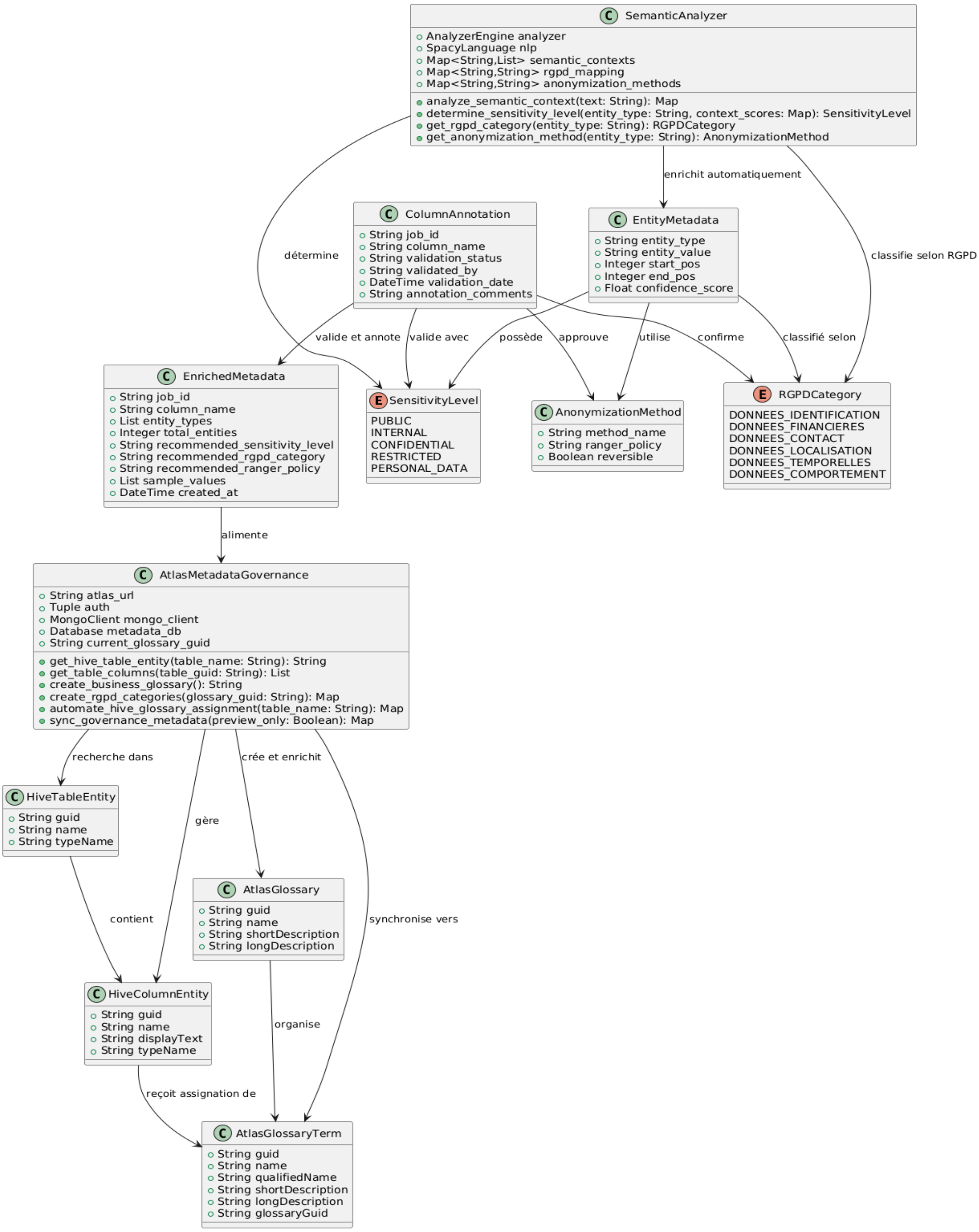

FIGURE 2.2 – Class Diagram - Data Governance and AI Integration

2.7 Sequence Diagrams

2.7.1 PII Detection Process with Semantic Enrichment

This sequence diagram illustrates the automatic detection and semantic enrichment process for sensitive data in CSV files.

The process follows a cell-by-cell analysis approach where the **SemanticAnalyzer** processes individual values and enriches them with GDPR classifications, sensitivity levels, and anonymization recommendations. The aggregation step consolidates cell-level analysis into column-level recommendations, which serve as input for AI-powered recommendations via Google Gemini.

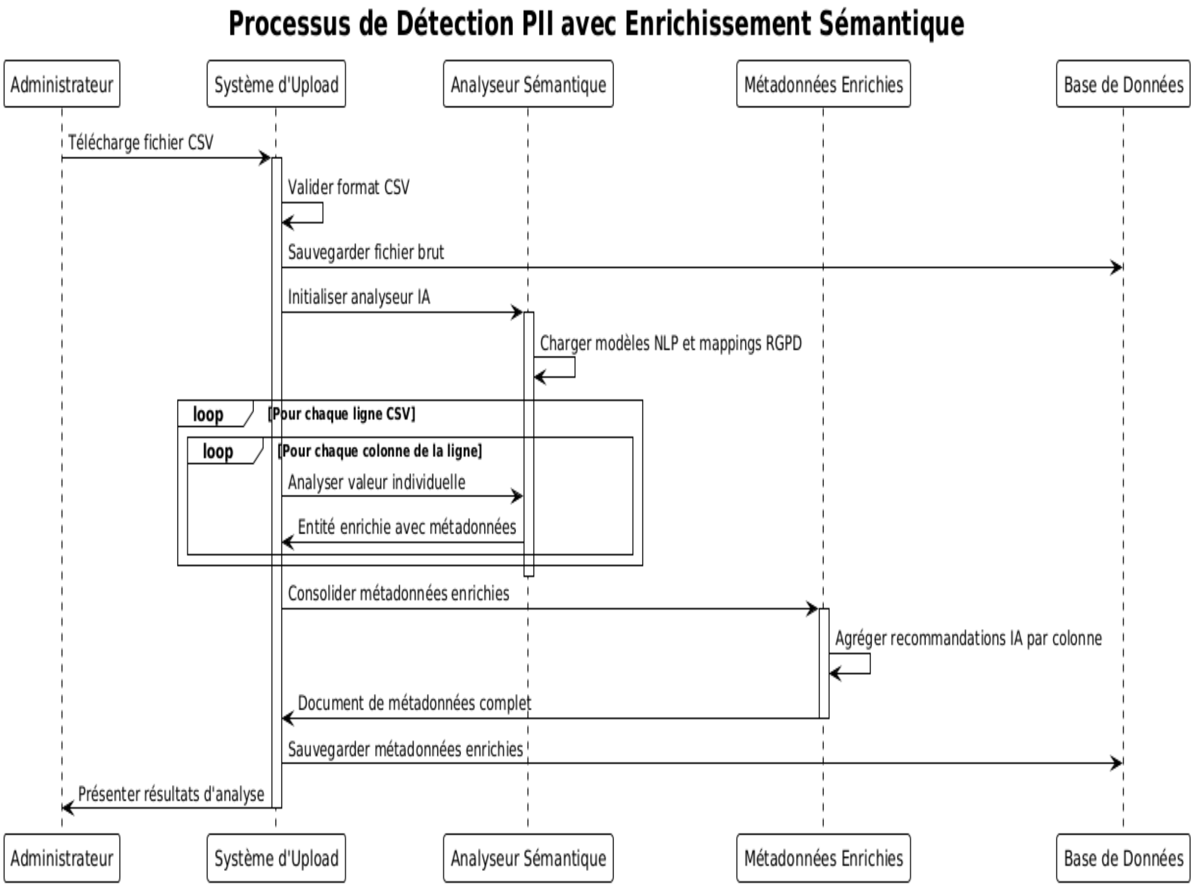

FIGURE 2.3 – PII Detection Process with Semantic Enrichment

2.7.2 Atlas Governance Synchronization Workflow

This sequence diagram shows the enterprise metadata governance workflow where validated metadata is synchronized to Apache Atlas. The workflow begins with data steward validation and proceeds through business glossary creation, GDPR category establishment, and automatic assignment of glossary terms to Hive table columns.

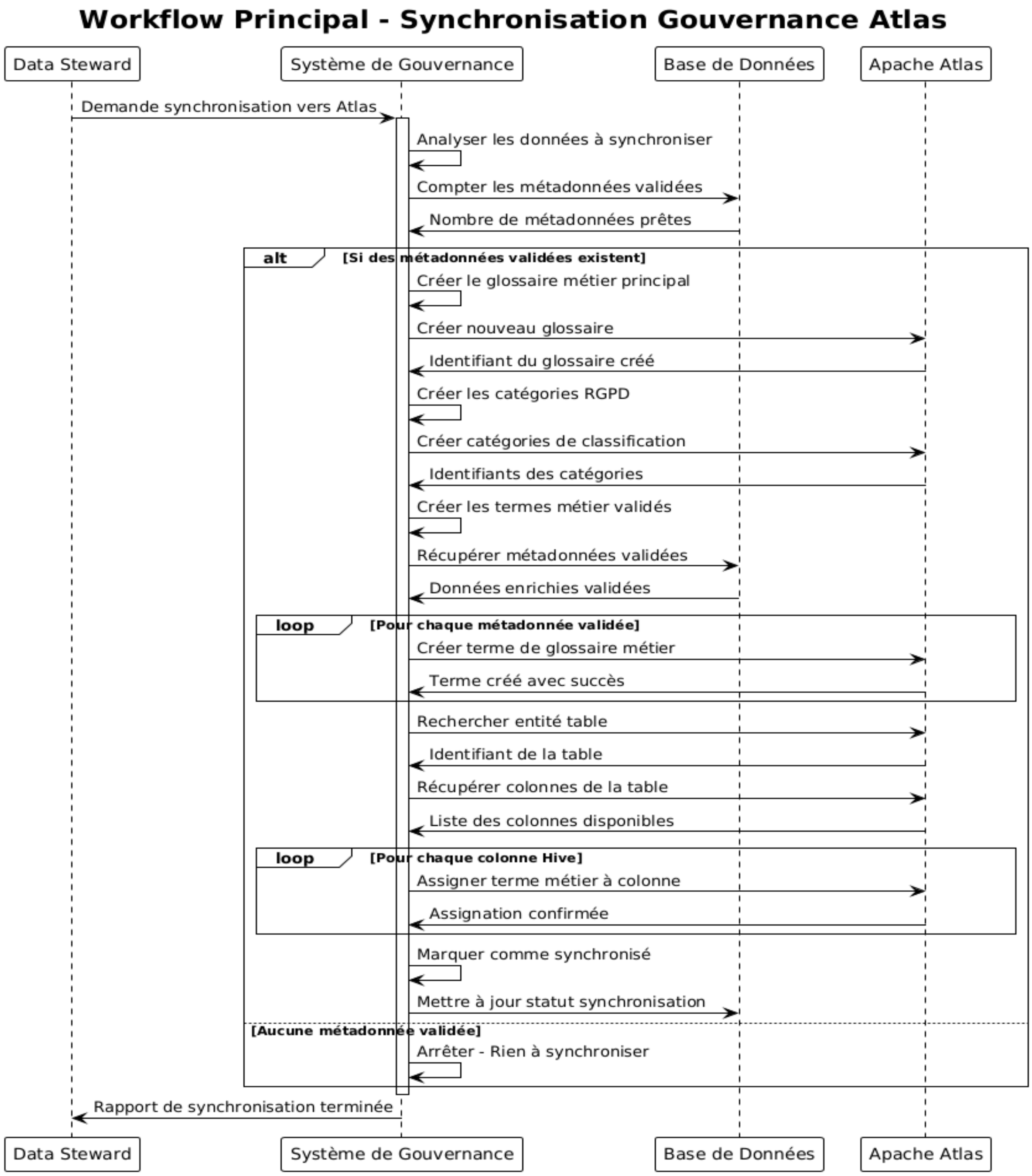

Figure 2.4 – Atlas Governance Synchronization Workflow

2.8 Design Patterns and Principles

The system follows several key design patterns :

- **Pipeline Pattern** : Sequential processing through detection → enrichment → aggregation → recommendation
- **Observer Pattern** : Automatic synchronization triggers upon metadata validation
- **Factory Pattern** : Dynamic creation of Atlas glossary terms, GDPR categories, and business glossaries based on validated metadata **Mapping Pattern** : Automatic assignment of glossary terms to existing Hive table columns

The aggregation step from cell-level analysis to column-level recommendations is critical as it transforms granular semantic analysis into structured dataset profiles that feed the AI recommendation engine. This enables Google Gemini to generate contextual recommendations for data governance and quality improvement.

2.9 Conclusion

This chapter has presented a comprehensive analysis and design framework for the automatic sensitive data detection and metadata governance platform. The proposed architecture successfully integrates cutting-edge AI technologies with enterprise-grade data governance capabilities, creating a robust solution for GDPR compliance and intelligent data stewardship.

Chapitre 3

Implementation and Technologies

Introduction

This section presents the technological choices adopted for developing the automatic sensitive data detection and metadata governance platform, highlighting their interaction and complementarity. We will detail the technologies used for both frontend and backend development, as well as the AI/ML components that power the intelligent data analysis capabilities.

The technology stack has been carefully selected to support enterprise-grade data governance workflows while maintaining scalability, security, and GDPR compliance requirements.

3.1 System Architecture Overview

Before diving into individual technologies and implementations, it is essential to understand how all components interact within the complete system architecture. Figure 3.1 presents the end-to-end workflow of the platform, illustrating the four major processing steps and the integration between different technological layers.

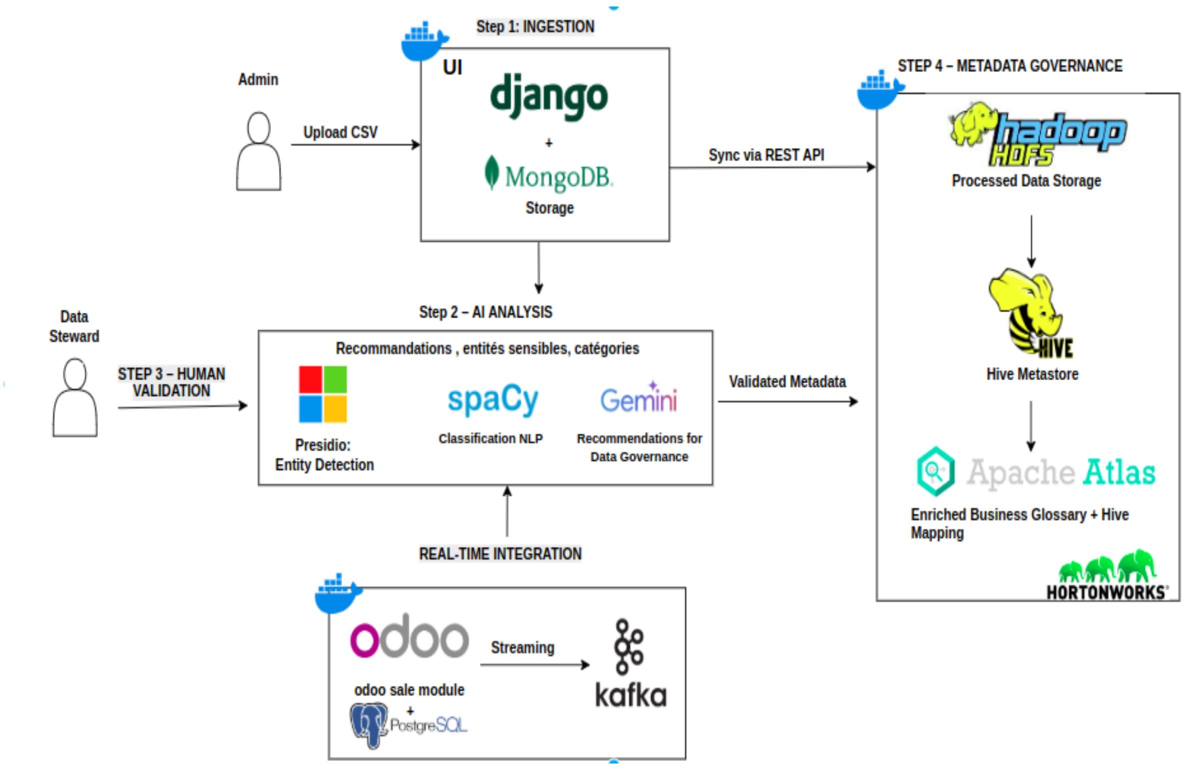

FIGURE 3.1 – Complete System Architecture - End-to-End Data Governance Workflow

3.1.1 Architecture Components and Data Flow

The system architecture implements a four-step workflow that processes sensitive data from ingestion to enterprise governance synchronization :

Step 1 : Data Ingestion

The platform supports two primary data ingestion methods :
- **CSV Upload (Admin) :** Administrators manually upload CSV files through the Django web interface. Files are stored in MongoDB using GridFS for efficient handling of large datasets.
- **Real-Time Streaming (Kafka) :** Customer data from Odoo ERP flows continuously through Apache Kafka topics. The system implements a batch-oriented consumer that buffers messages (50 records or 5-minute intervals) before processing.

Both ingestion methods converge into the Django + MongoDB storage layer, where raw data is persisted in document collections optimized for subsequent AI analysis.

Step 2 : AI Analysis

The core intelligence layer processes ingested data through three integrated AI/ML components :
- **Microsoft Presidio :** Performs entity detection using pre-built and custom recognizers to identify PII types (PERSON, EMAIL_ADDRESS, PHONE_NUMBER, ID_MAROC, IBAN_CODE, etc.). Presidio provides high-confidence entity recognition with configurable anonymization policies.
- **spaCy NLP :** Enhances detection capabilities through semantic analysis and contextual understanding. The French language models enable lemmatization, part-of-speech tagging, and intelligent classification beyond simple pattern matching.
- **Google Gemini API :** Generates intelligent recommendations across four governance categories (COMPLIANCE, SECURITY, QUALITY, GOVERNANCE). Gemini analyzes dataset profiles and provides actionable insights for data stewards.

The AI analysis produces enriched metadata including detected entity types, RGPD categories, sensitivity levels, and Apache Ranger policy recommendations for each column.

Step 3 : Human Validation

The platform implements a human-in-the-loop validation workflow where data stewards review and approve AI-generated metadata recommendations :
- **Metadata Review Interface :** Data stewards access a comprehensive dashboard displaying detected entities, recommended RGPD categories, sensitivity classifications, and suggested Ranger policies.
- **Validation Modal :** Each column can be validated, rejected, or modified through an interactive modal interface. Stewards can accept AI recommendations or apply domain expertise to override automated classifications.
- **Quality Gates :** Only validated metadata (with `ready_for_atlas_sync` flag set to true) proceeds to enterprise synchronization, ensuring governance quality.

This validation step ensures that automated AI insights are reviewed by domain experts before being propagated to enterprise data catalogs.

Step 4 : Metadata Governance Synchronization

Validated metadata is synchronized to the Hortonworks Data Platform (HDP) ecosystem through REST API integration :

- **Apache Hive Integration :** The system maps validated columns to existing Hive table structures, enabling data stewards to apply governance policies at the enterprise data warehouse level.
- **Apache Atlas Business Glossary :** The platform automatically creates glossary terms, assigns them to RGPD category glossaries, applies sensitivity classifications, and maps terms to Hive columns. This eliminates manual glossary management in the Atlas interface.

The HDP integration bridges the gap between AI-generated insights and actionable enterprise data governance policies.

3.1.2 Real-Time Integration Layer

The architecture includes a dedicated real-time processing pipeline for operational data :

- **Odoo ERP Source :** Customer data originates from Odoo's sale module, including customer information (name, email, phone, location) and order details (reference, amount, date).
- **PostgreSQL Change Data Capture :** Odoo uses PostgreSQL as its backend database. Changes to customer records trigger events that are captured and published to Kafka topics.
- **Apache Kafka Streaming :** The `odoo-customer-data` topic carries real-time customer events. The Kafka consumer processes batches and applies the same AI analysis pipeline used for CSV data.
- **Consolidated Metadata View :** The system aggregates metadata across multiple Kafka batches, enabling data stewards to validate column definitions once for all streaming data.

This real-time integration ensures continuous GDPR compliance monitoring for operational business data as it flows through enterprise systems.

3.1.3 Technology Layer Interactions

The architecture demonstrates clear separation of concerns across technology layers :

- **Presentation Layer (UI) :** Django templates with Tailwind CSS provide role-based interfaces for administrators (CSV upload, user management) and data stewards (metadata validation, quality assessment).
- **Application Layer (Django) :** Handles authentication, session management, business logic orchestration, and REST API endpoints for Atlas synchronization.
- **Storage Layer (MongoDB) :** Document database stores raw CSV data (GridFS), enriched metadata, AI recommendations, and Kafka job results.
- **AI/ML Layer :** Modular components (Presidio, spaCy, Gemini) operate independently, enabling technology substitution without affecting other layers.

— **Enterprise Integration Layer (HDP) :** Apache Atlas and Hive provide the bridge between AI-generated insights and enterprise data governance infrastructure.

3.1.4 Architectural Patterns and Design Principles

The system implements several key architectural patterns :

— **Batch-Oriented Processing :** Optimizes database operations by processing data in chunks (1000-row CSV batches, 50-message Kafka batches).
— **Event-Driven Architecture :** Kafka enables decoupled, asynchronous processing of real-time operational data.
— **Microservices Principles :** The Kafka consumer runs as an isolated Docker container, enabling independent scaling and deployment.
— **Human-in-the-Loop AI :** Balances automation with human oversight, ensuring AI recommendations are validated before enterprise synchronization.
— **API-First Design :** REST APIs enable integration with external systems (Atlas, Gemini) and future extensibility.

This architectural foundation enables the platform to handle enterprise-scale data governance while maintaining flexibility, scalability, and GDPR compliance requirements. The following sections detail the specific technologies and implementations that realize this architecture.

3.2 Technologies

The Technology section focuses on the tools, technologies, and frameworks used for developing the web application. It explains the rationale behind technological choices, their integration within the project, and their impact on system effectiveness and performance.

3.2.1 Software Tools

Git : Git is a distributed version control system used to manage changes in software projects. It enables tracking modifications, reverting changes when necessary, and collaborating effectively with other developers. Essential for maintaining code integrity and managing the complex multi-component architecture of our AI-powered data governance system.

FIGURE 3.2 – Git

GitHub : GitHub is the leading collaborative platform for open-source development, built on Git. It enables project hosting, public code contribution, and collaboration management through pull requests. Its tools facilitate teamwork, issue tracking, and workflow automation. Indispensable today, it serves as the reference ecosystem for developers working on enterprise-scale projects.

FIGURE 3.3 – GitHub

Docker and Docker Compose : Docker provides containerization technology that ensures consistent deployment across different environments. Docker Compose orchestrates multi-service applications, enabling seamless integration between the Django web application, MongoDB database, and external AI services. This containerized approach simplifies deployment and ensures reproducible environments for data governance workflows.

FIGURE 3.4 – Docker

Playwright :
Playwright is an end-to-end testing framework developed by Microsoft that enables automated testing of web applications across multiple browsers, including Chromium, Firefox, and WebKit. It provides powerful APIs to simulate user interactions such as clicking, typing, and navigating between pages, ensuring that the user interface behaves as expected. In this project, Playwright was used to perform functional and regression tests on the developed interfaces, guaranteeing their stability and consistency after each update. Its integration into the development workflow enhances software reliability and supports continuous integration practices.

FIGURE 3.5 – Playwright

3.2.2 Programming Languages

Programming languages enable the implementation of technical solutions for detecting, processing, and protecting sensitive data within enterprise governance frameworks.

Python : Python is an interpreted, versatile, and open-source programming language designed for productivity and code readability. It combines the power of object-oriented, imperative, and functional programming paradigms with clear and intuitive syntax, making it an ideal tool for implementing GDPR-compliant solutions and AI/ML pipelines for sensitive data detection.

FIGURE 3.6 – Python

3.2.3 Web Development Frameworks

Web frameworks provide pre-built software structures to efficiently implement GDPR requirements and data governance workflows. They accelerate development while natively integrating functionalities dedicated to data protection, traceability, and risk management.

Django :
Django is a full-stack web framework in Python, designed for rapid development of robust and secure applications. Its "batteries-included" structure makes it an ideal choice for implementing a GDPR-compliant backend, combining productivity with adherence to data protection requirements. The framework provides built-in security features, ORM capabilities, and scalable architecture patterns essential for enterprise data governance systems.

FIGURE 3.7 – Django

HTML, CSS, JavaScript, and Tailwind CSS :
The user interface has been developed using standard web technologies : **HTML**, **CSS**, and **JavaScript**, enabling content structuring, component styling, and application interactivity. To accelerate visual design and ensure a modern, responsive interface, we utilized the **Tailwind CSS** framework, which offers a utility-first approach to design, facilitating rapid prototyping and style harmonization throughout the project.

FIGURE 3.8 – Tailwind CSS

3.2.4 Artificial Intelligence and Machine Learning Technologies

The AI/ML stack forms the core intelligence layer of the system, enabling automatic detection, semantic analysis, and intelligent recommendations for sensitive data governance.

Microsoft Presidio :
Microsoft Presidio is an open-source framework for PII detection and anonymization. It provides pre-built recognizers for common entity types (PERSON, EMAIL_ADDRESS, PHONE_NUMBER) and supports custom recognizers for region-specific entities like Moroccan ID cards (ID_MAROC) and IBAN codes. Presidio serves as the primary detection engine in our system, offering high-confidence entity recognition with configurable anonymization policies.

FIGURE 3.9 – Microsoft Presidio

spaCy Natural Language Processing :
spaCy is an industrial-strength NLP library that powers the semantic analysis capabilities of our system. The SemanticAnalyzer component leverages spaCy's French language models to understand contextual meaning, perform lemmatization, and extract semantic features from detected entities. This enables intelligent classification of data sensitivity levels and GDPR category mapping beyond simple pattern matching.

FIGURE 3.10 – spaCy NLP

Google Gemini API :
Google Gemini API provides advanced generative AI capabilities for intelligent recommendation generation. The GeminiClient component interfaces with Gemini's large language models to analyze dataset profiles and generate contextual recommendations for data quality improvement, security enhancement, and GDPR compliance. This AI integration enables the system to provide human-like insights for complex data governance decisions.

FIGURE 3.11 – Google Gemini

3.2.5 Database Technologies

The system implements a hybrid database architecture optimized for different data types and access patterns required by AI processing and enterprise governance workflows.

MongoDB :
MongoDB serves as the primary document database for storing CSV data, enriched metadata, and AI-generated recommendations. Its flexible schema design accommodates the dynamic nature of metadata enrichment and supports complex nested structures required for entity detection results. The system uses multiple MongoDB databases : `csv_anonymizer_db` for raw and processed data, and `metadata_validation_db` for governance workflows.

FIGURE 3.12 – MongoDB

3.2.6 Enterprise Data Governance Integration

The system integrates with enterprise-grade data governance platforms to provide comprehensive metadata management and compliance workflows.

Apache Atlas :
Apache Atlas provides enterprise metadata management and data governance capabilities. The AtlasMetadataGovernance component synchronizes validated metadata to Atlas, creating business glossaries, GDPR categories, and glossary terms. This integration enables automatic assignment of metadata terms to Hive table columns, bridging the gap between AI-generated insights and enterprise data catalogs.

FIGURE 3.13 – Apache Atlas

Apache Hive :
Apache Hive serves as the data warehouse metadata catalog, providing structured access to enterprise data tables and columns. The system maps detected sensitive data entities to existing Hive table structures, enabling data stewards to apply governance policies at the enterprise level. Integration with Hive ensures that AI-generated metadata recommendations are actionable within existing data infrastructure.

FIGURE 3.14 – Apache Hive

3.2.7 Real-Time Data Integration and Enterprise Systems

The system integrates with enterprise ERP systems through real-time streaming infrastructure, enabling continuous data governance for operational business data.

Apache Kafka :

Apache Kafka serves as the distributed streaming platform for real-time data ingestion from enterprise systems. The system implements a batch-oriented consumer that buffers incoming messages and processes them in batches of 50 records or every 5 minutes. This batching strategy optimizes database operations while maintaining near-real-time data governance capabilities. Kafka integration enables the platform to automatically detect and classify sensitive data as it flows through operational systems, ensuring continuous GDPR compliance monitoring.

FIGURE 3.15 – Apache Kafka

Odoo ERP :

Odoo is the enterprise resource planning system that serves as the primary data source for real-time governance workflows. The system consumes customer data from Odoo's sale module through Kafka topics, including customer information (name, email, phone, location) and order details (reference, amount, date).

FIGURE 3.16 – Odoo ERP

3.3 Implementation and User Interfaces

3.3.1 Authentication and User Management

The authentication system provides secure access control and user management capabilities for the platform, implementing session-based authentication with MongoDB as the user credential store.

Login Interface

The login interface serves as the entry point for all users, implementing session-based authentication with role-based redirection. The authentication workflow validates user credentials against MongoDB's users collection using Django's password hashing mechanisms , then establishes a session and redirects users to role-appropriate dashboards.

Upon successful authentication, administrators are redirected to the CSV upload interface, while data stewards access their specialized metadata validation dashboard.

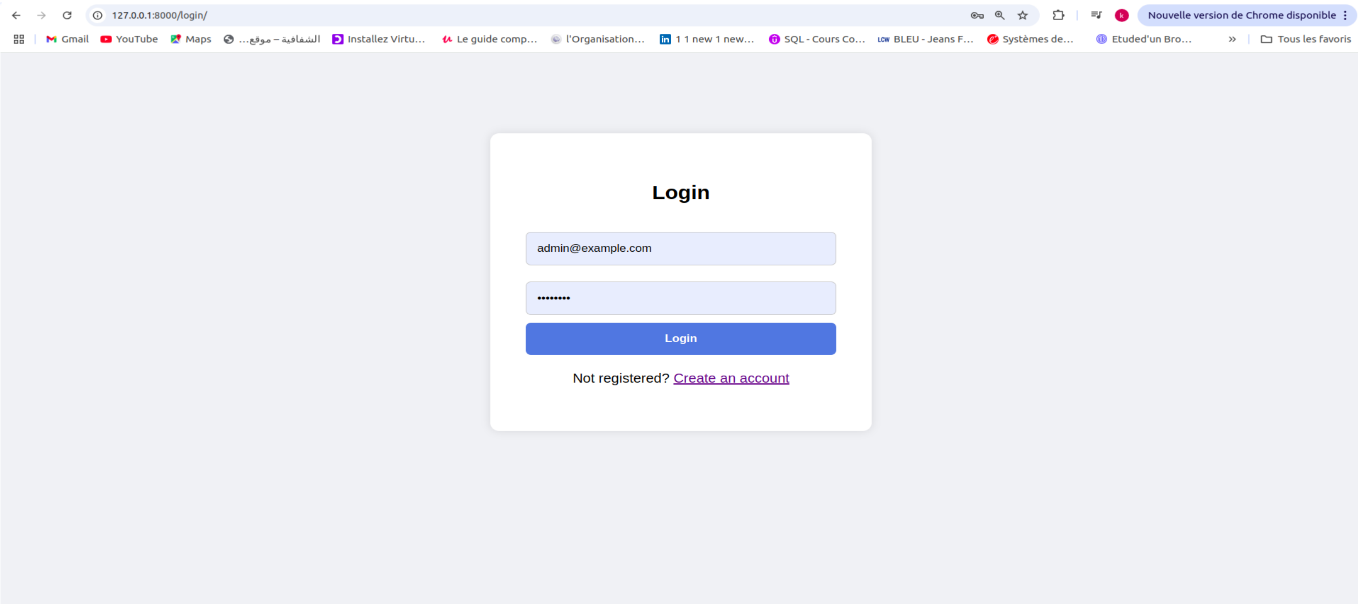

FIGURE 3.17 – Login Interface - Session-Based Authentication Entry Point

3.3.2 Administrator Interfaces

Administrators have access to comprehensive system management capabilities including user management, CSV file processing, entity detection configuration, and system-wide analytics.

CSV File Upload Interface

The upload interface serves as the main entry point for the automatic sensitive data detection system. This interface, accessible only to administrators, allows uploading CSV files that will be analyzed by artificial intelligence to identify sensitive personal data.

The upload workflow implements several key technical features :

— **Streaming File Processing :** Large CSV files are processed in chunks to avoid memory overflow, using Django's `file.chunks()` iterator to handle files of arbitrary size.

— **GridFS Storage :** Uploaded files are stored in MongoDB's GridFS system, enabling efficient storage and retrieval of large binary objects while maintaining metadata associations.

— **Chunk-Based Data Storage :** CSV data is divided into 1000-row chunks , optimizing query performance for large datasets during entity detection and metadata enrichment.

The interface provides real-time feedback on upload progress and automatically initiates the Presidio-based entity detection pipeline upon successful file storage.

FIGURE 3.18 – CSV File Upload Interface - Streaming Upload with GridFS Storage

User Administration Interface

The administrative interface provides complete CRUD (Create, Read, Update, Delete) operations for user management, accessible only to users with admin role. This interface enables administrators to manage the platform's user base, assign roles, and control access to sensitive data governance workflows.

The user table displays comprehensive information including name, email, role, and action buttons for modification and deletion.

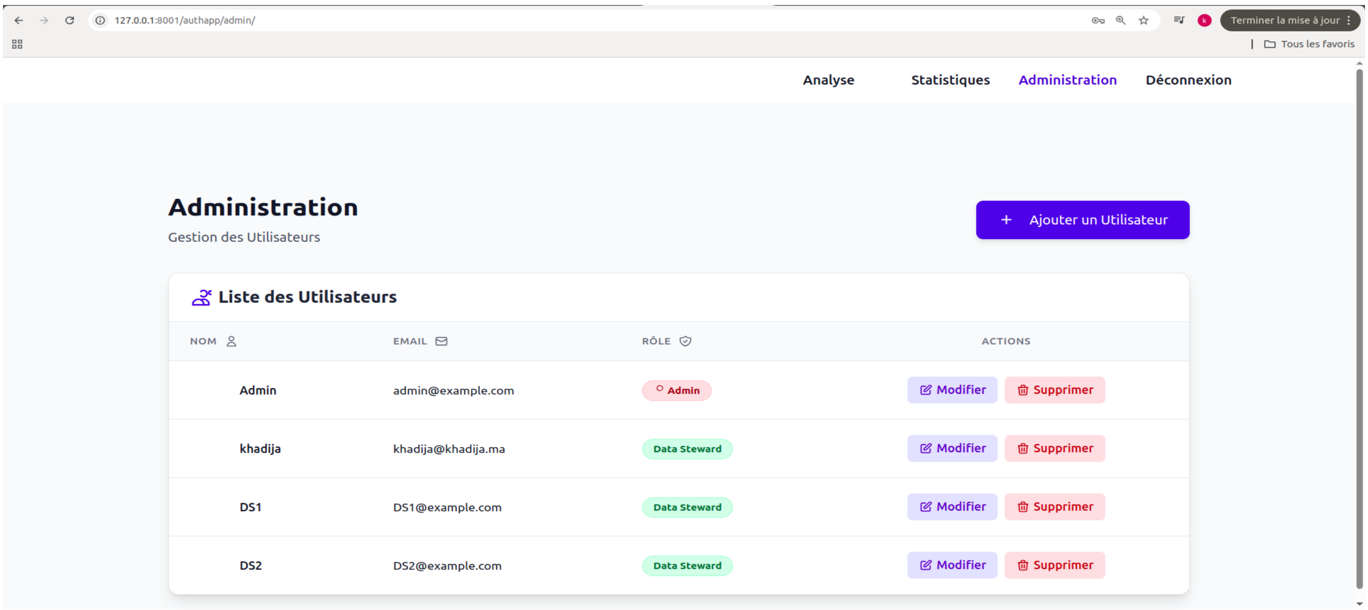

FIGURE 3.19 – User Management Interface - CRUD Operations with Role-Based Access Control

Sensitive Entity Detection and Selection Interface

The entity selection interface allows administrators to configure anonymization settings after CSV upload and AI analysis. This interface presents the results of Presidio's entity detection analysis and enables selective anonymization based on business requirements.

The interface workflow operates as follows :

- **Automatic Entity Detection :** Upon CSV upload, the system analyzes column samples using the IntelligentAutoTagger which combines Presidio's recognizers with spaCy's semantic analysis to detect entity types including PERSON, EMAIL_ADDRESS, PHONE_NUMBER, CREDIT_CARD, IBAN_CODE, ID_MAROC, and LOCATION.
- **Selective Anonymization :** Administrators can choose which entity types to anonymize by selecting checkboxes, allowing preservation of certain data types for analytical purposes while protecting truly sensitive information.
- **Column-Level Analysis :** The system employs stratified random sampling to analyze column content, selecting representative samples (5-10% of data) distributed across the beginning, middle, and end sections of each column. This adaptive sampling strategy ensures statistical representativeness while maintaining computational efficiency, enabling accurate column-to-entity type mapping regardless of data distribution patterns.
- **Masking Strategy :** Selected entities are replaced with "[MASQUÉ]" tokens during anonymization, using Presidio's AnonymizerEngine with custom operator configurations.

The interface also provides a direct link to view AI-generated recommendations for data governance, enabling administrators to understand the compliance implications of detected entities before proceeding with anonymization.

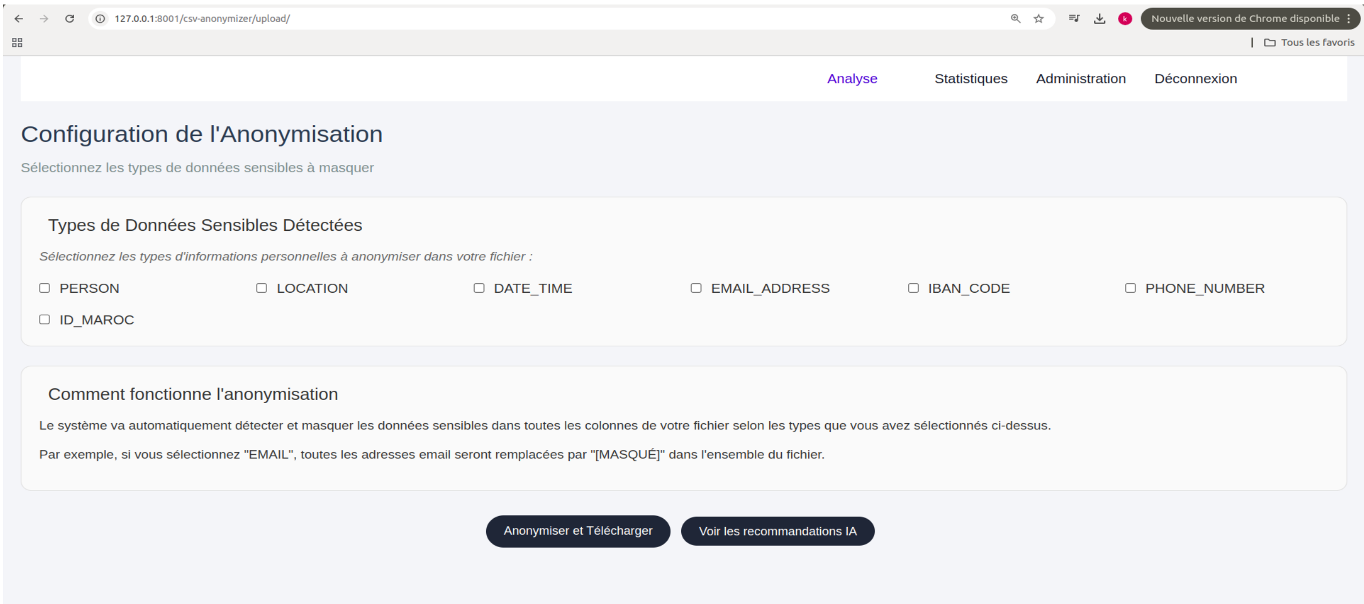

FIGURE 3.20 – Entity Type Selection Interface - Configurable Anonymization with Presidio Integration

AI-Generated Recommendations Interface

The recommendations interface displays AI-generated insights for data governance and quality improvement, leveraging Google Gemini's large language models to analyze dataset profiles and generate actionable recommendations.

The recommendation system implements a sophisticated multi-category analysis framework :

- **Four Enterprise Categories :** Recommendations are organized into COMPLIANCE (RGPD/GDPR requirements), SECURITY (encryption and access controls), QUALITY (data completeness and consistency), and GOVERNANCE (metadata documentation) categories.
- **Priority-Based Scoring :** Each recommendation receives a priority score (1-10) and confidence level (0-1), with critical issues (priority ≥ 8) highlighted in red, warnings (6-7.9) in yellow, and informational items (<6) in green.
- **Tabbed Navigation :** The interface uses JavaScript-powered tabs to switch between categories, with each tab displaying category-specific recommendations and immediate action items.
- **Intelligent Fallback :** If Gemini API fails or times out, the system generates rule-based recommendations using dataset profile analysis, ensuring recommendations are always available.

The overall governance score (displayed prominently at the top) aggregates individual category scores to provide an at-a-glance assessment of dataset compliance and quality status.

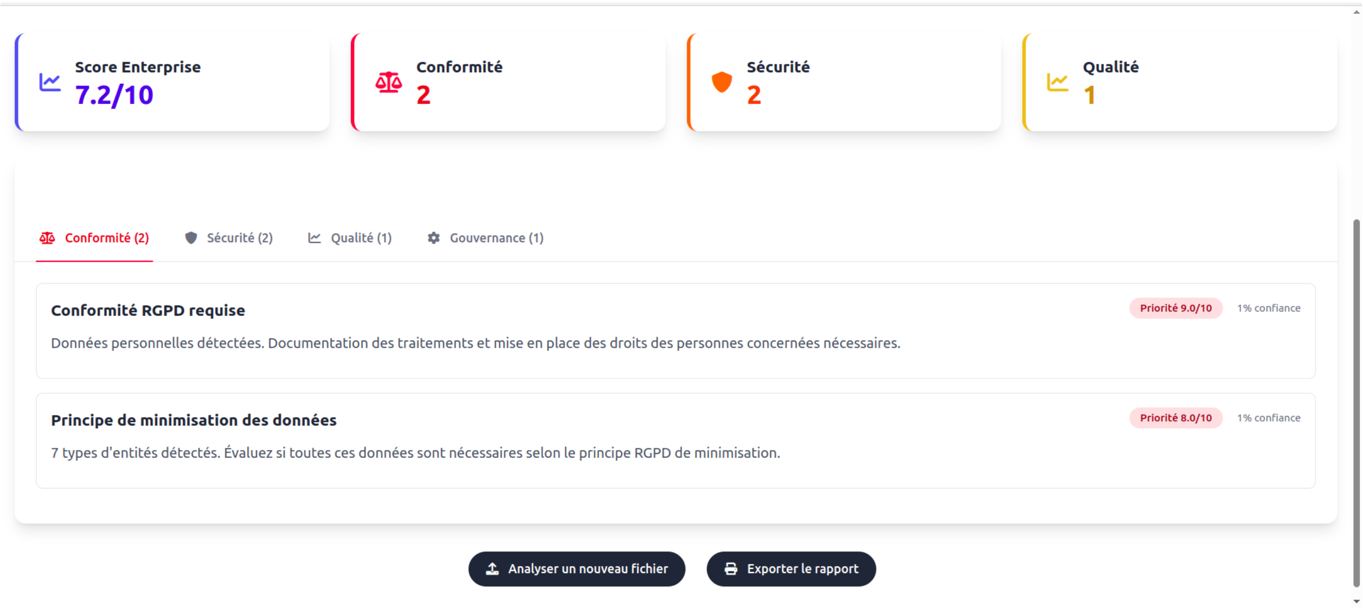

FIGURE 3.21 – AI Recommendations Dashboard - Compliance Category

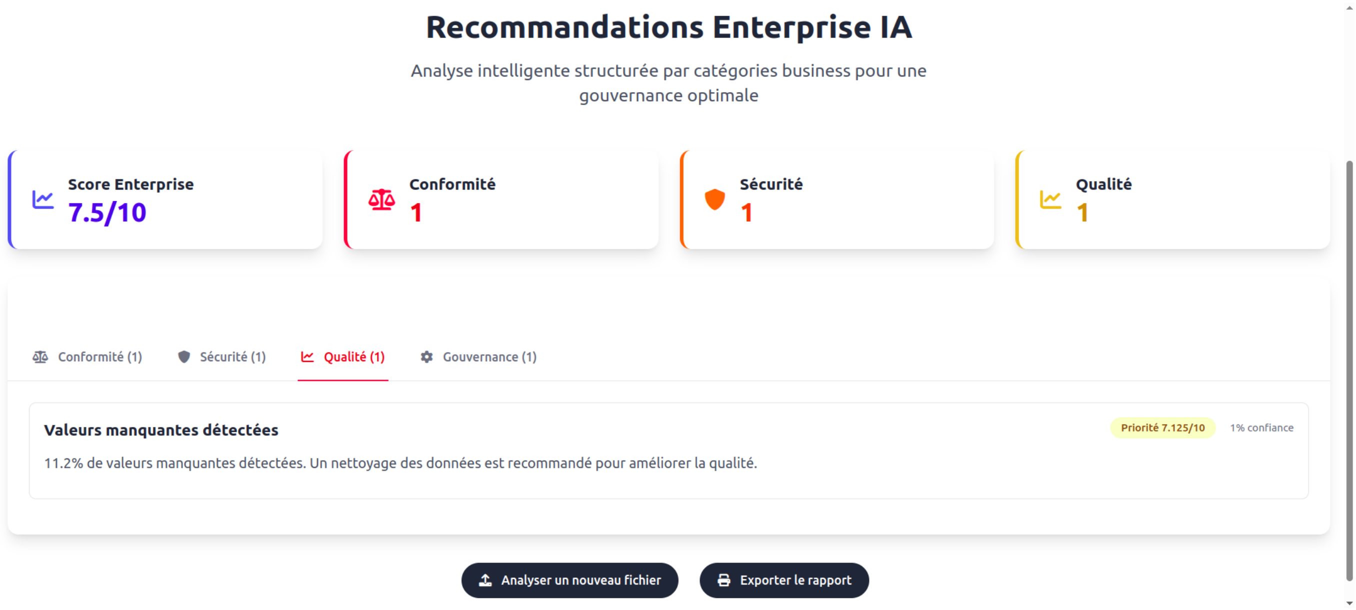

FIGURE 3.22 – AI Recommendations Dashboard - Quality Category

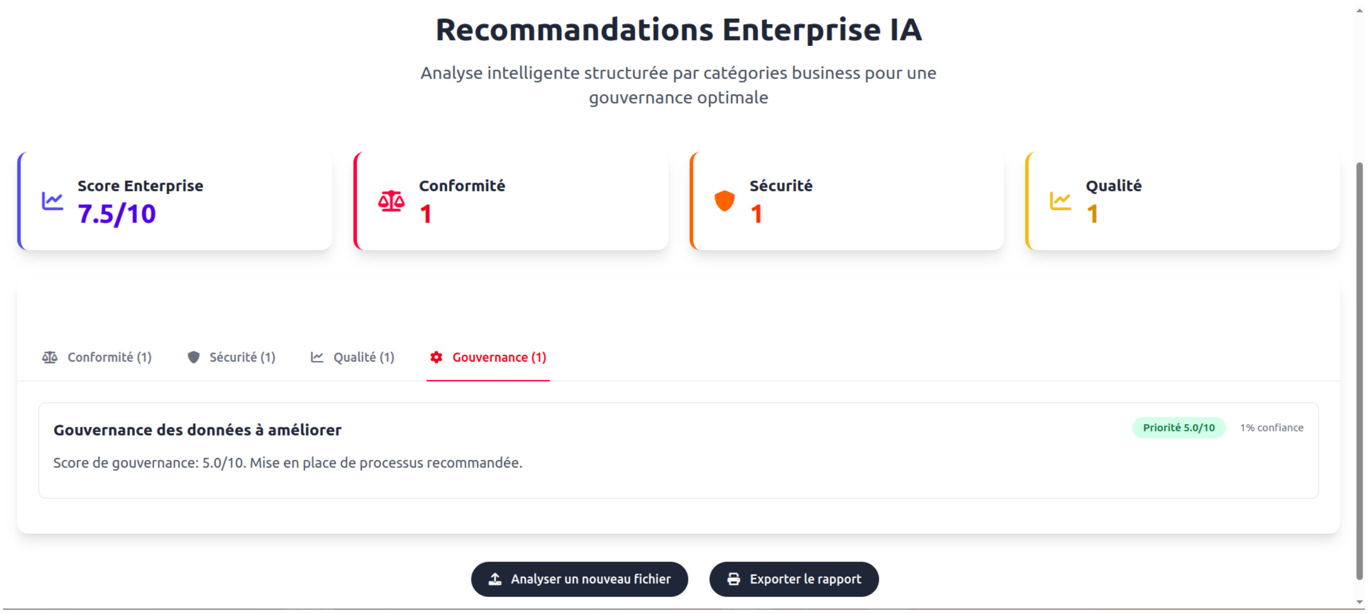

FIGURE 3.23 – AI Recommendations Dashboard - Governance Category

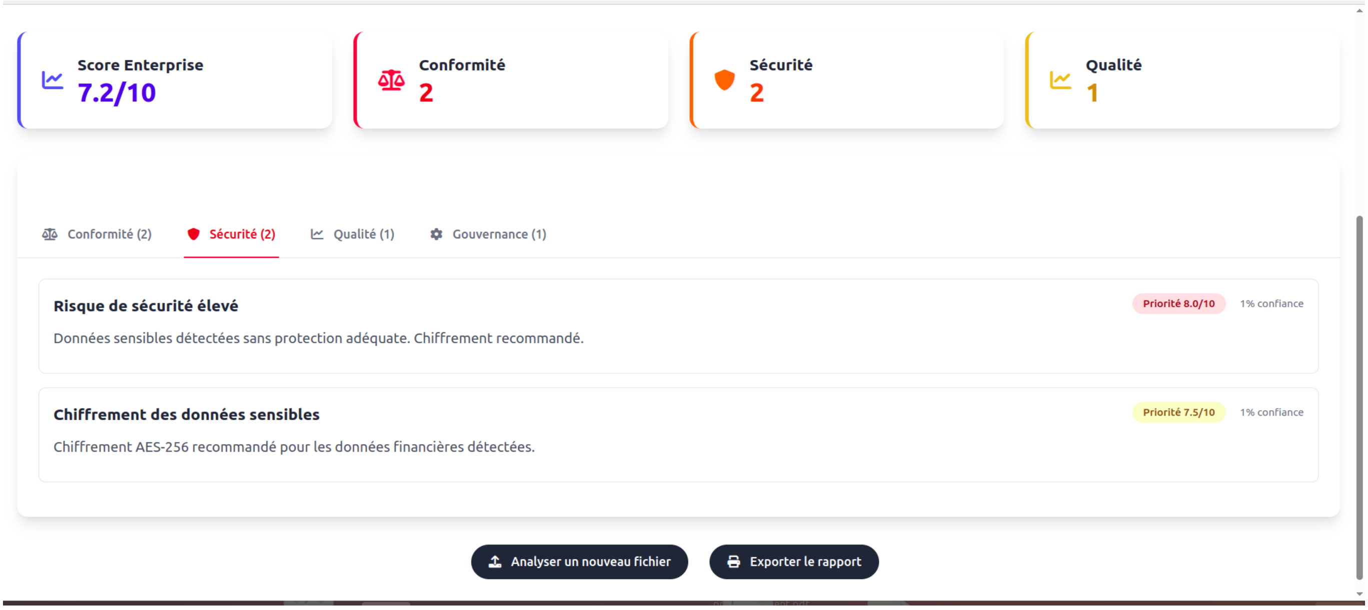

FIGURE 3.24 – AI Recommendations Dashboard - Security Category

3.4 Data Steward Interfaces

Data stewards have specialized interfaces for metadata validation, data quality assessment, and governance workflows.

3.4.1 Enriched Metadata Management Interface and Automatic Business Glossary Generation

The metadata interface displays enriched column analysis that directly feeds into automated business glossary generation in Apache Atlas. This interface represents the core of the human-in-the-loop validation workflow, where data stewards review AI-generated metadata recommendations before they are synchronized to the enterprise data catalog.

The interface implements a comprehensive metadata enrichment pipeline :

— **Automatic Entity Detection :** The system analyzes each column using the IntelligentAutoTagger, which combines Microsoft Presidio's recognizers with spaCy's semantic analysis to detect entity types (PERSON, EMAIL_ADDRESS, PHONE_NUMBER, ID_MAROC, IBAN_CODE, etc.).

— **RGPD Category Assignment :** Each column receives an automatic RGPD category recommendation based on detected entity types, following a priority cascade : identification data (PERSON, ID_MAROC), financial data (IBAN_CODE, CREDIT_CARD), contact data (PHONE_NUMBER, EMAIL_ADDRESS), and location data.

— **Sensitivity Level Classification :** The system assigns sensitivity levels (PERSONAL_DATA, RESTRICTED, CONFIDENTIAL, INTERNAL, PUBLIC) based on the privacy impact of detected entities.

— **Ranger Policy Recommendations :** Each column receives a recommended Apache Ranger masking policy (full masking, partial masking, hashing, encryption, generalization) appropriate for the detected entity type.

— **Validation Status Tracking :** Each column displays its validation status (pending, validated, rejected) with color-coded badges, enabling data stewards to track their progress through the validation workflow.

The validation workflow uses a modal dialog interface where data stewards can review and modify AI recommendations before approval. The modal pre-fills all fields with AI-generated recommendations, allowing stewards to accept them as-is or make adjustments based on domain expertise. Upon validation, the system sets the `ready_for_atlas_sync` flag to true, marking the metadata as eligible for synchronization to Apache Atlas.

When data stewards validate columns through the annotation workflow, the system automatically creates comprehensive glossary terms in Apache Atlas, assigns them to appropriate RGPD categories, applies sensitivity classifications, and maps them to Hive columns - all without requiring manual glossary management in the Atlas interface.

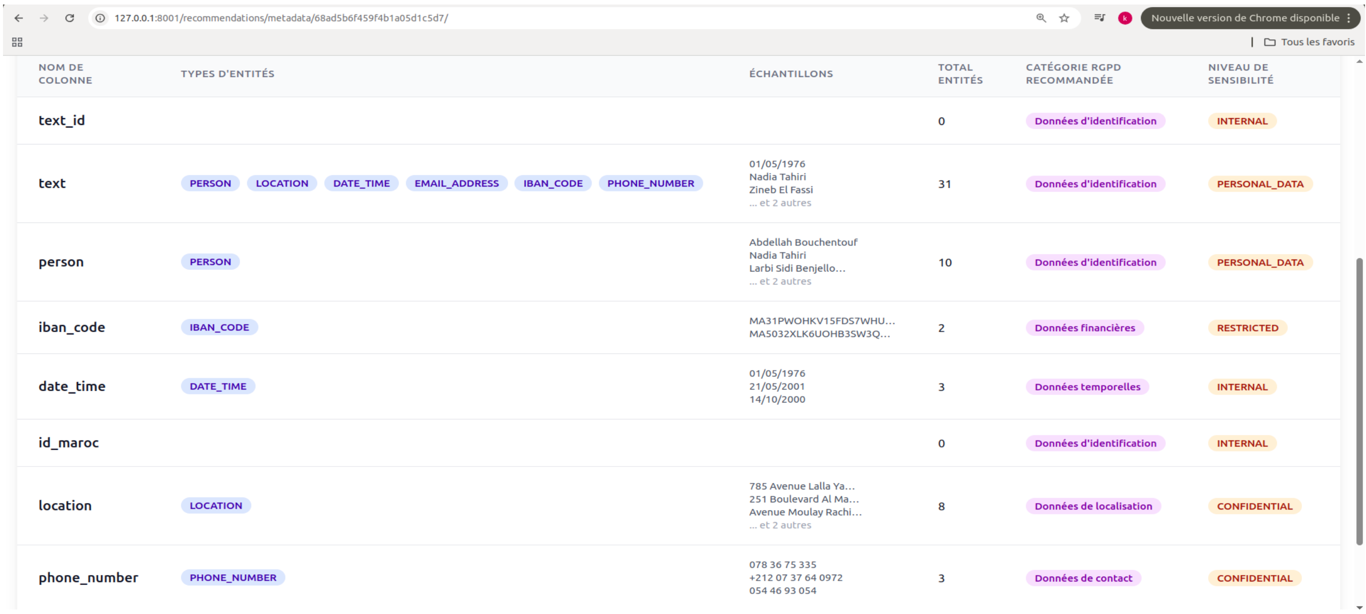

FIGURE 3.25 – Enriched Metadata Validation Interface - Column-Level Analysis with AI Recommendations

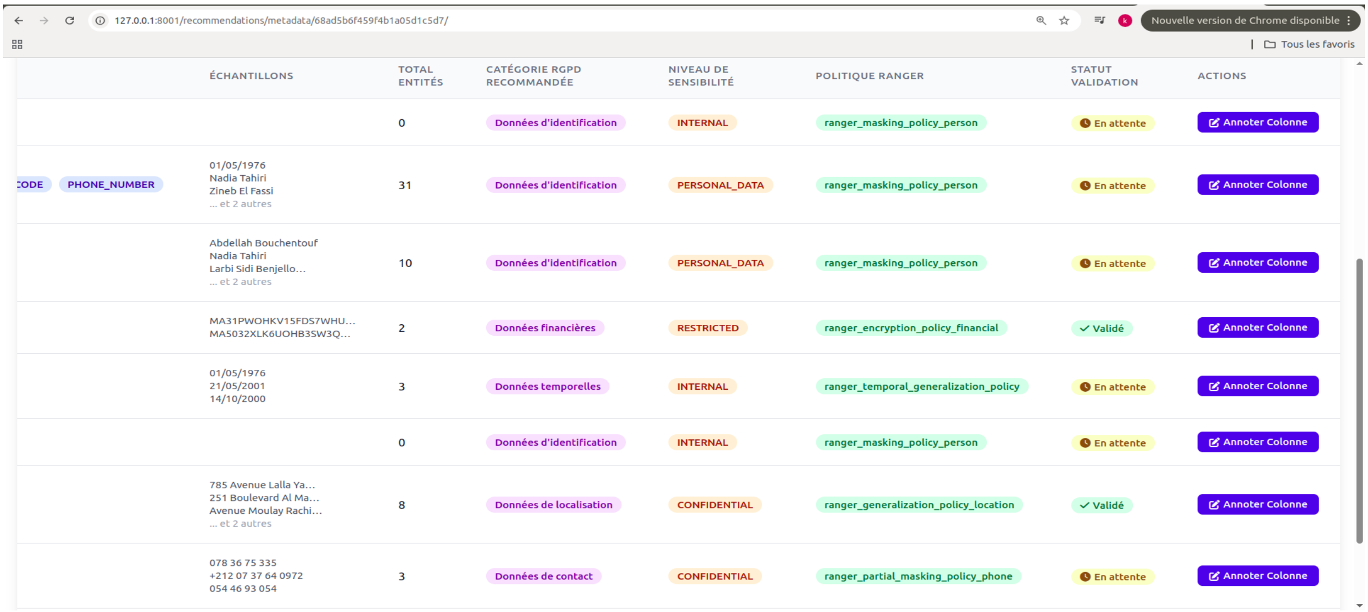

FIGURE 3.26 – Metadata Validation Modal - Human-in-the-Loop Review Workflow

3.4.2 Apache Atlas Business Glossary - Automated Metadata Synchronization

The automatic synchronization between validated metadata and Apache Atlas represents the culminating phase of the data governance workflow, transforming AI-generated insights into an enterprise-grade business glossary. This section demonstrates the technical implementation and business value of the automated glossary generation system.

Glossary Term Structure and Professional Descriptions

Figure 3.27 illustrates a complete glossary term entry for the `PERSON_TERM`, demonstrating the comprehensive metadata enrichment achieved through the synchronization process. Each term follows a standardized structure designed to meet both regulatory compliance requirements and operational data governance needs.

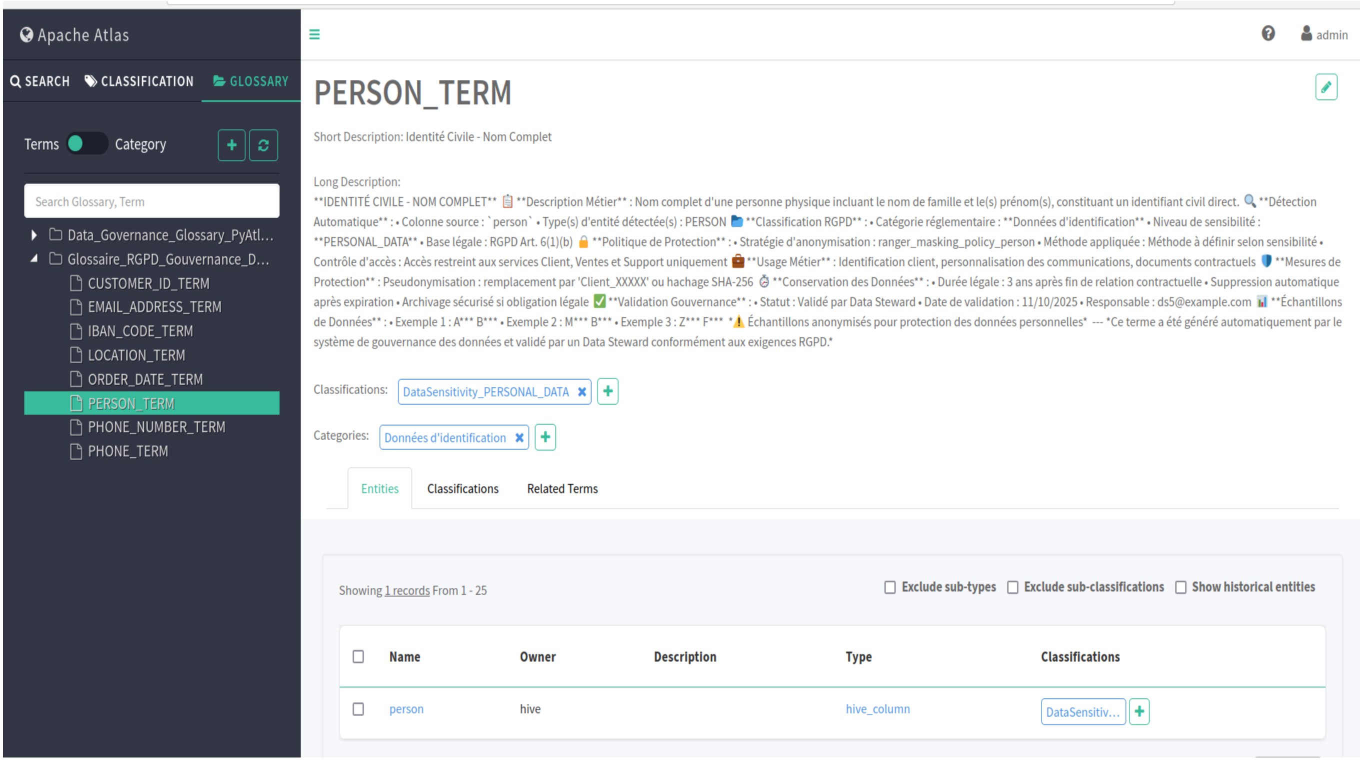

FIGURE 3.27 – PERSON_TERM in Apache Atlas - Complete Glossary Entry with RGPD Classification

The term structure includes several critical components :

Business-Oriented Nomenclature The system transforms technical column names (`person`, `phone_number`, `iban_code`) into semantically meaningful business terms. For instance, the short description **"Identité Civile - Nom Complet"** (Civil Identity - Full Name) provides immediate business context that technical stakeholders, data stewards, and compliance officers can all understand. This bridging of technical and business vocabularies is essential for cross-functional data governance.

Regulatory-Compliant Long Descriptions The long description section demonstrates the automated generation of comprehensive, RGPD-compliant documentation. Key elements include :

— **Automatic Detection Metadata :** Traceability information showing the source column (`person`), detected entity types (PERSON), and detection confidence rate. This enables data quality auditing and verification of AI-generated classifications.

— **RGPD Classification Framework :** Automatic assignment to regulatory categories ("Données d'identification") with explicit legal basis (RGPD Art. 6(1)(b) - Contract execution). This mapping ensures that every data element has a documented legal justification for processing, satisfying accountability requirements under GDPR Art. 5(2).

— **Protection Policies :** Concrete recommendations for anonymization strategies (pseudonymization : replacement by 'Client_XXXXX' or SHA-256 hashing) and access controls (restricted to Client Services, Sales, and Support teams only). These

operational guidelines enable immediate implementation of privacy-by-design principles.
— **Business Usage Context :** Clear articulation of legitimate use cases (customer identification, communication personalization, contractual documents) to prevent function creep and ensure purpose limitation compliance (RGPD Art. 5(1)(b)).
— **Data Retention Policy :** Specific retention period (3 years after end of contractual relationship) based on French/Moroccan legal requirements, with automatic deletion after expiration. This operationalizes storage limitation principles (RGPD Art. 5(1)(e)).
— **Governance Validation Metadata :** Complete audit trail including validation status, validation date (11/10/2025), and responsible data steward (ds5@example.com). This creates accountability and enables compliance audits by supervisory authorities (CNIL/CNDP).
— **Anonymized Data Samples :** Representative examples (A*** B***, Exemple 2 : M*** P***) that illustrate data format while respecting data minimization principles. The automatic anonymization of samples within the glossary itself demonstrates privacy-by-design at the metadata layer.

Entity Relationship - Hive Column Mappings

The **Entities** tab (visible in Figure 3.27) demonstrates the automatic assignment of glossary terms to physical Hive table columns. The table shows :
— **Name :** `person` - The actual Hive column name in the data warehouse
— **Owner :** `hive` - The technical ownership within the Hadoop ecosystem
— **Type :** `hive_column` - The Atlas entity type
— **Classifications :** `DataSensitivity_PERSONAL_DATA` - The security classification automatically propagated from the term to the column

This automatic mapping creates **operational data lineage** where :

1. A data steward validates a CSV column in the Django interface
2. The system creates a glossary term in Atlas with comprehensive descriptions
3. The term is automatically assigned to the corresponding Hive table column
4. Security classifications flow from the term to the physical column

This end-to-end automation eliminates the traditional bottleneck where data governance teams must manually document and classify every column in enterprise data warehouses. The system reduces glossary creation time from weeks to minutes while ensuring consistency and completeness.

Classification System - Sensitivity-Based Access Controls

Figure 3.28 demonstrates the Apache Atlas classification system, which enables automated policy enforcement through integration with Apache Ranger. Classifications serve as security labels that trigger access control policies across the entire data platform.

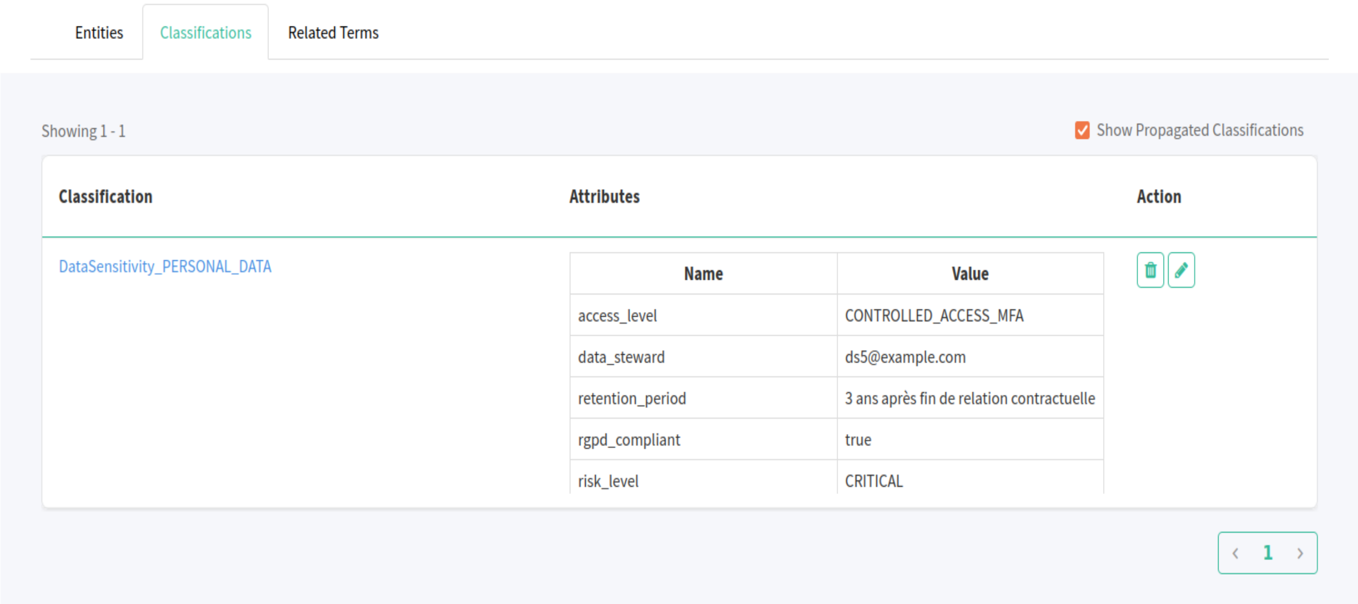

FIGURE 3.28 – DataSensitivity_PERSONAL_DATA Classification - Automated Security Controls

The `DataSensitivity_PERSONAL_DATA` classification illustrated includes five critical attributes :

1. **access_level : CONTROLLED_ACCESS_MFA** - Mandates multi-factor authentication for any access to columns bearing this classification. This implements technical security measures required under RGPD Art. 32(1)(a).
2. **data_steward : ds5@example.com** - Designates the accountable data steward responsible for this data category, establishing clear ownership for governance decisions and compliance inquiries.
3. **retention_period : 3 ans après fin de relation contractuelle** - Embeds retention policy directly in the metadata layer, enabling automated lifecycle management and demonstrating storage limitation compliance (RGPD Art. 5(1)(e)).
4. **rgpd_compliant : true** - Boolean flag indicating that this classification has been validated for RGPD compliance, facilitating compliance reporting and audit workflows.
5. **risk_level : CRITICAL** - Risk-based classification that enables prioritization of security controls and incident response procedures. Critical-risk data elements receive the highest level of monitoring and protection.

This classification system enables **policy-based data governance** where security controls are automatically applied based on data sensitivity rather than requiring manual configuration for each table or column.

Comparison with Manual Approaches

Table 3.1 compares the automated approach with traditional manual glossary management :

TABLE 3.1 – Automated vs Manual Glossary Management Comparison

Criterion	Manual Approach	Automated System
Time per 50 columns	20-30 hours	5 minutes (post-validation)
Description quality	Inconsistent, varies by author	Standardized, comprehensive
Legal references	Often missing or incorrect	Automatic, always current
Retention periods	Manual research required	Auto-calculated by category
Audit trail	Limited or absent	Complete (validator, date)
Hive integration	Manual, error-prone	Automatic via REST API
Classification propagation	Manual per-column	Automatic term-to-column
Scalability	Linear cost growth	Fixed cost, unlimited scale
Human error rate	15-20% (typos, wrong categories)	<2% (only validation errors)

This automated approach represents a paradigm shift from "glossaries as documentation artifacts" to "glossaries as operational governance infrastructure" where metadata actively drives security controls, compliance reporting, and data discovery workflows.

Future Enhancements

Potential extensions to the glossary system include :

— **Multi-language Support :** Generate descriptions in French, English, and Arabic simultaneously to support multilingual teams and international compliance requirements.

— **Version Control :** Track changes to term definitions over time, enabling historical compliance audits ("what was our classification policy in Q2 2024 ?").

— **Impact Analysis :** When a data steward proposes changing a term's sensitivity level, automatically calculate downstream impacts (which Ranger policies would change, which users would lose access).

— **Glossary-Driven Data Discovery :** Enable business users to search "find all datasets containing customer financial data" and receive results based on glossary classifications rather than technical column names.

3.4.3 Data Quality Management Interface

The data quality interface provides tools for assessing and improving dataset quality through automated analysis and manual corrections. This interface leverages the DataQualityEngine component, which uses Google Gemini's AI capabilities to analyze data completeness, consistency, and identify duplicates.

The quality assessment workflow operates as follows :

— **Completeness Analysis :** The system calculates the percentage of non-null values for each column, identifying columns with missing data that may impact analytical accuracy.

— **Consistency Checking :** The engine detects format inconsistencies, data type mismatches, and value range violations across columns, providing specific examples of problematic records.

— **Duplicate Detection :** The system identifies exact duplicate rows and near-duplicate records based on configurable similarity thresholds.

— **Interactive Cleaning Actions :** Data stewards can select from three automated cleaning operations : remove duplicates, remove rows with missing values, or fix inconsistencies through standardization.

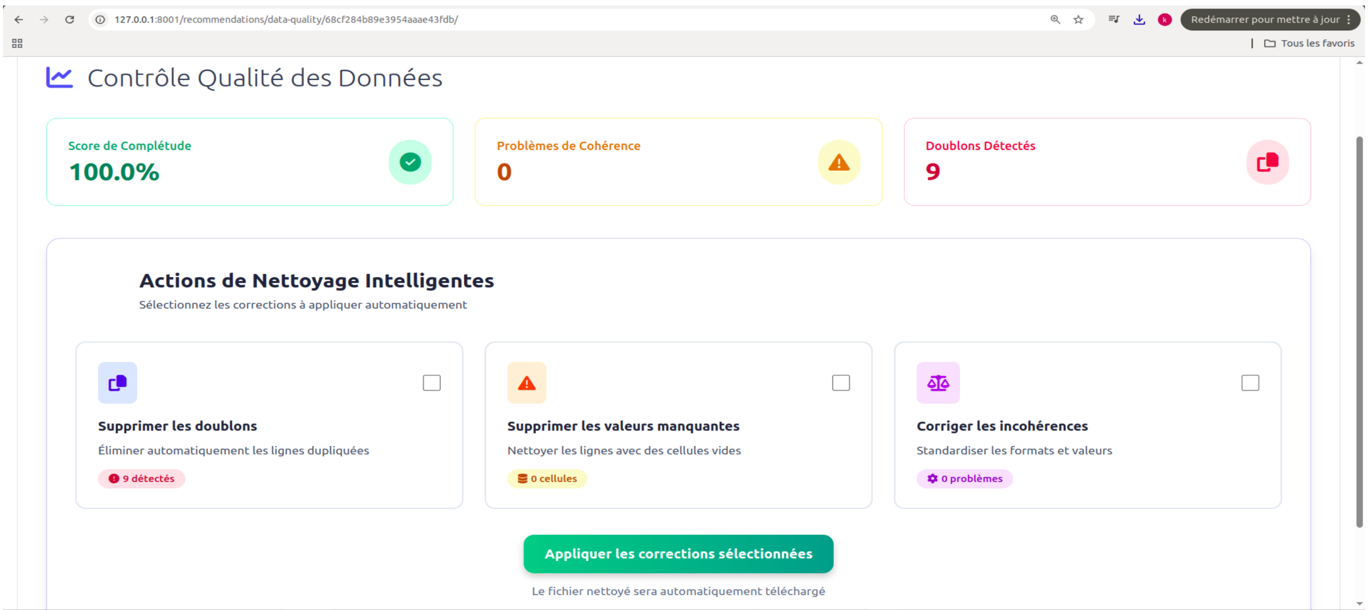

FIGURE 3.29 – Data Quality Assessment Interface - Quality Metrics Dashboard

3.4.4 Real-Time Data Integration from Odoo ERP

The system implements real-time data governance for operational business data through Apache Kafka streaming integration with Odoo ERP. This integration enables continuous monitoring and classification of sensitive customer data as it flows through enterprise systems, ensuring GDPR compliance for live operational data.

Kafka Consumer Architecture

The Kafka consumer implements a batch-oriented processing strategy optimized for database efficiency while maintaining near-real-time governance capabilities. The consumer operates as follows :

— **Message Buffering :** Incoming messages from the `odoo-customer-data` topic are buffered in memory until reaching 50 records or a 5-minute timeout, whichever occurs first.

— **Batch Processing :** When the buffer threshold is reached, all accumulated messages are processed as a single job.

— **Standardized Schema :** Each Kafka message contains some standardized fields from Odoo's sale module : `customer_id`, `name`, `email`, `phone`, `location`, `order_reference`, `order_amount`, and `order_date`...

The consumer runs as a dedicated Docker container, separate from the main Django web application, ensuring isolation and independent scalability.

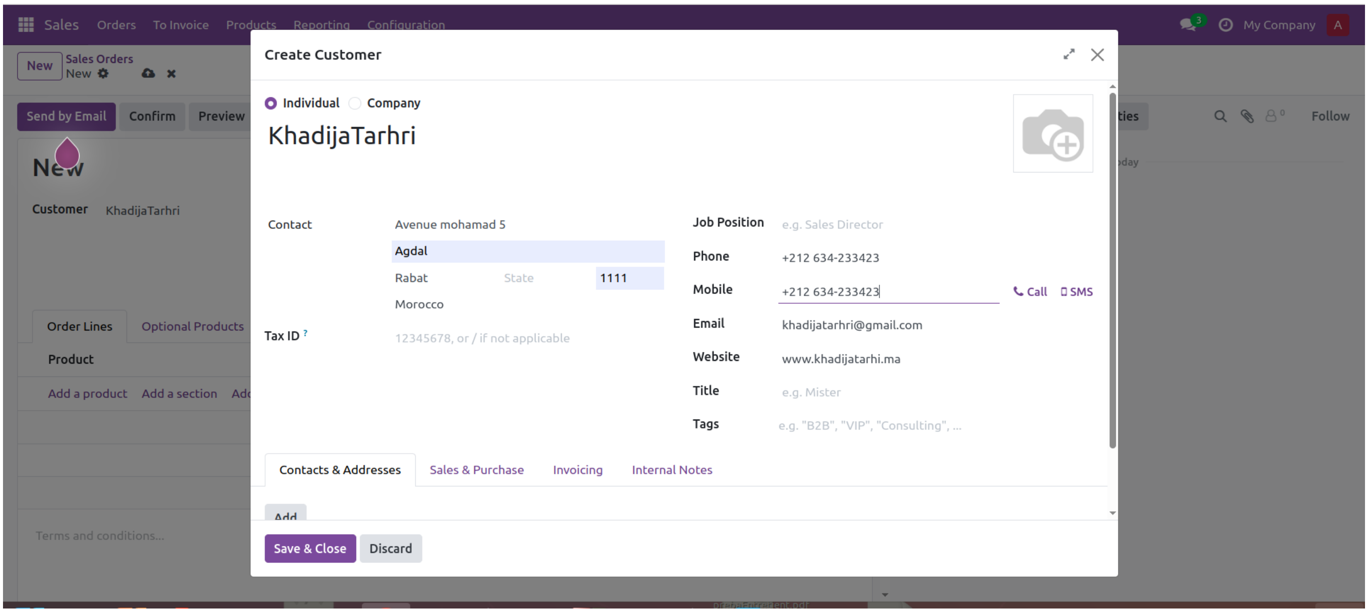

FIGURE 3.30 – Odoo Sale Module - Customer Creation Interface Generating Kafka Events

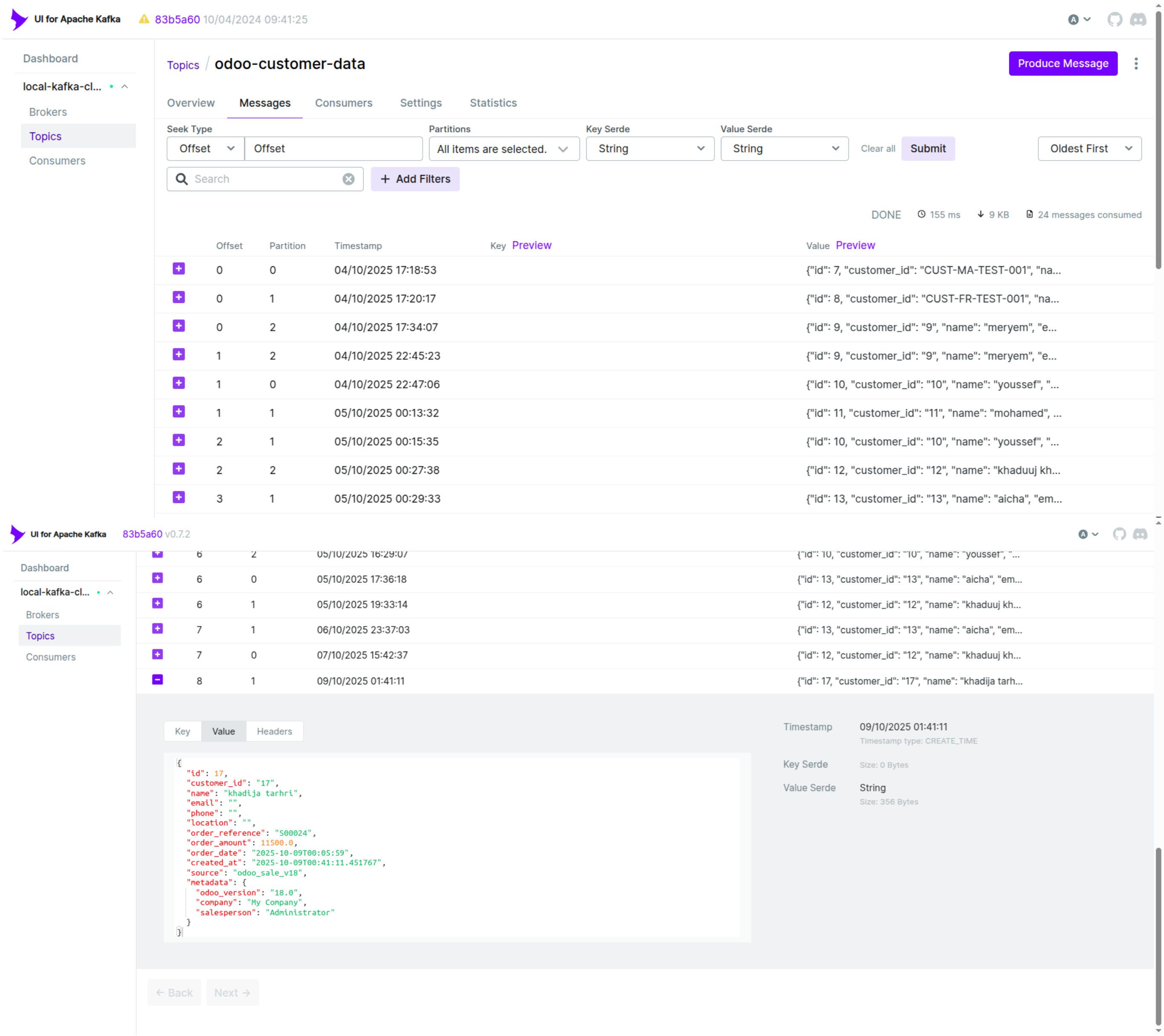

FIGURE 3.31 – Kafka UI - Real-Time Messages in odoo-customer-data Topic

3.4.5 Odoo Metadata Governance Interface

The Odoo metadata interface provides a consolidated view of all customer data ingested from Kafka streams, aggregating metadata across multiple batches to enable efficient validation workflows. This interface addresses the unique challenge of streaming data governance, where the same column types appear across multiple Kafka batches over time.

The consolidation architecture implements several sophisticated features :

- **Semantic Analysis Integration :** Each column is analyzed using the SemanticAnalyzer component, which provides context-aware entity detection with confidence scores, going beyond simple pattern matching to understand semantic meaning.
- **Unified Validation Workflow :** Data stewards can validate a column , and the validation automatically applies to all batches containing that column, ensuring consistent governance policies across streaming data.

— **Real-Time Statistics Dashboard :** The interface displays live metrics including total columns analyzed, total records processed, validation status distribution (validated, pending, rejected), and average confidence scores across all batches.
— **Advanced Filtering Capabilities :** The interface provides real-time search by column name and filtering by validation status.

The interface processes customer data from Odoo's sale module with 8 standard fields : `customer_id`, `name`, `email`, `phone`, `location`, `order_reference`, `order_amount`, and `order_date`. The Kafka consumer implements a batch-oriented architecture that buffers incoming messages and flushes them as a single job when the buffer reaches 50 messages or 5 minutes have elapsed, optimizing database operations while maintaining near-real-time governance capabilities.

The validation modal for Odoo metadata includes all standard governance fields (entity type, validation status, RGPD category, sensitivity level, Ranger policy, and comments), with the critical difference that validation updates are applied to ALL metadata records across ALL Kafka batches containing that column name, not just a single job.

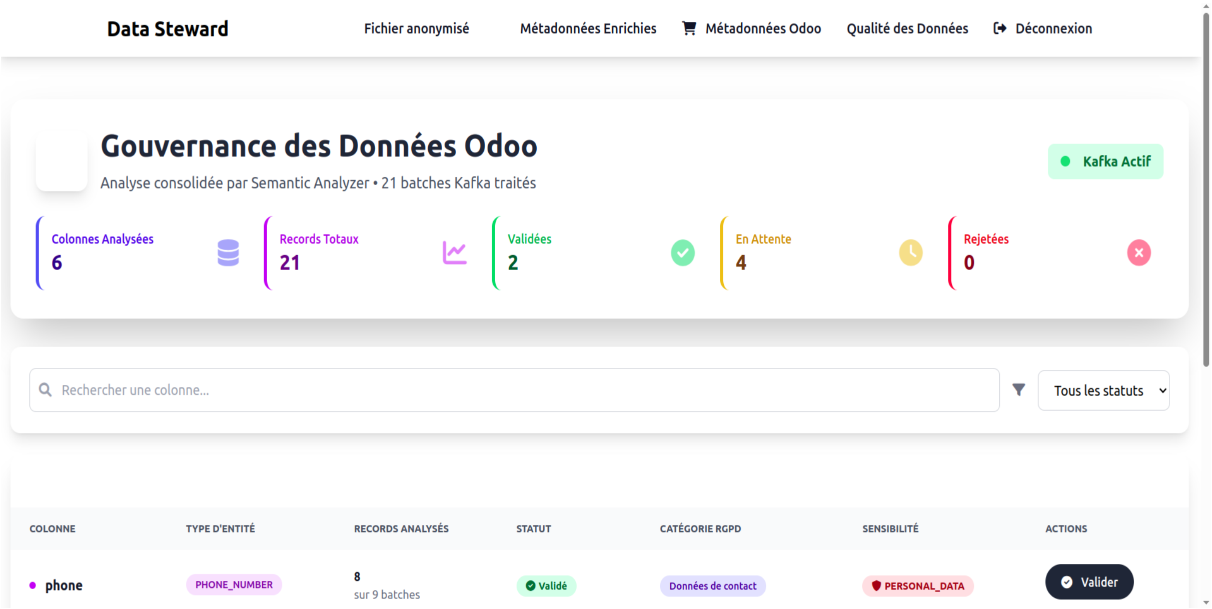

FIGURE 3.32 – Odoo Consolidated Metadata Interface - Statistics Dashboard

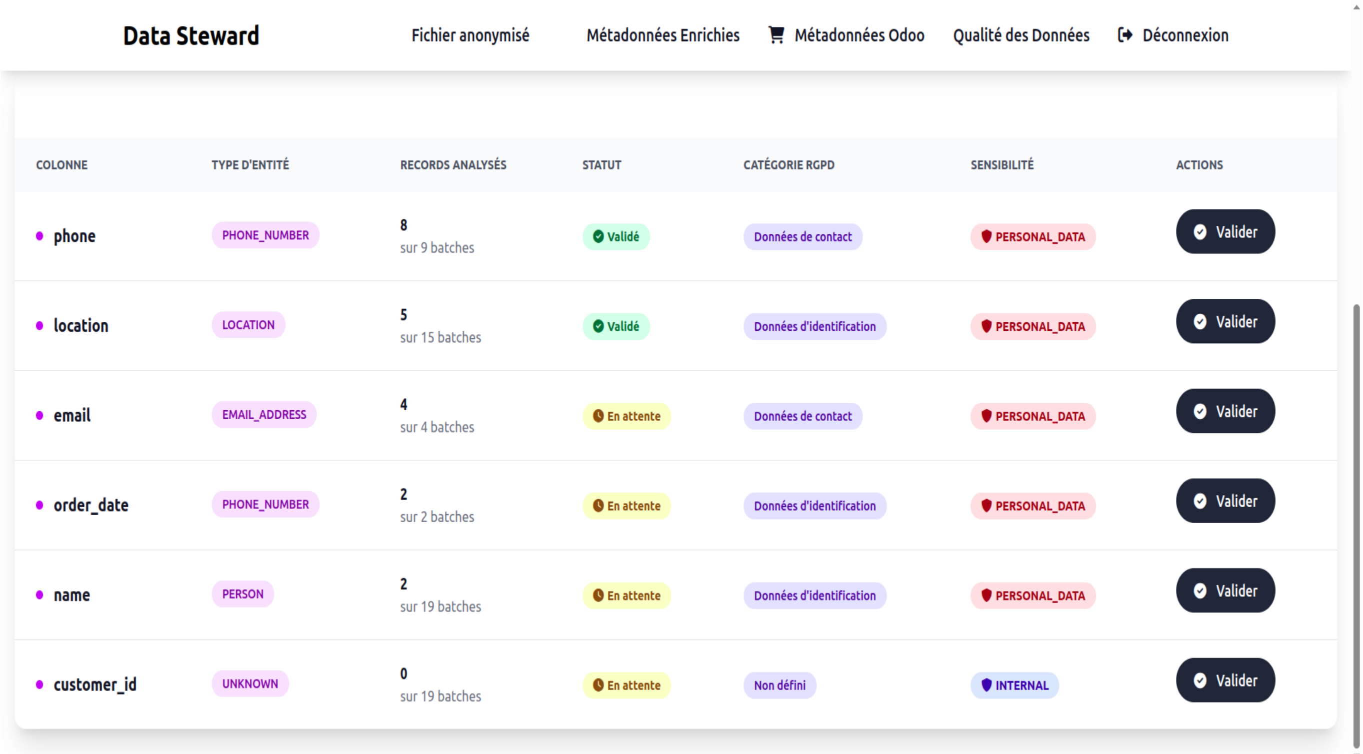

FIGURE 3.33 – Odoo Metadata Table - Column-Level Analysis with Batch Consolidation

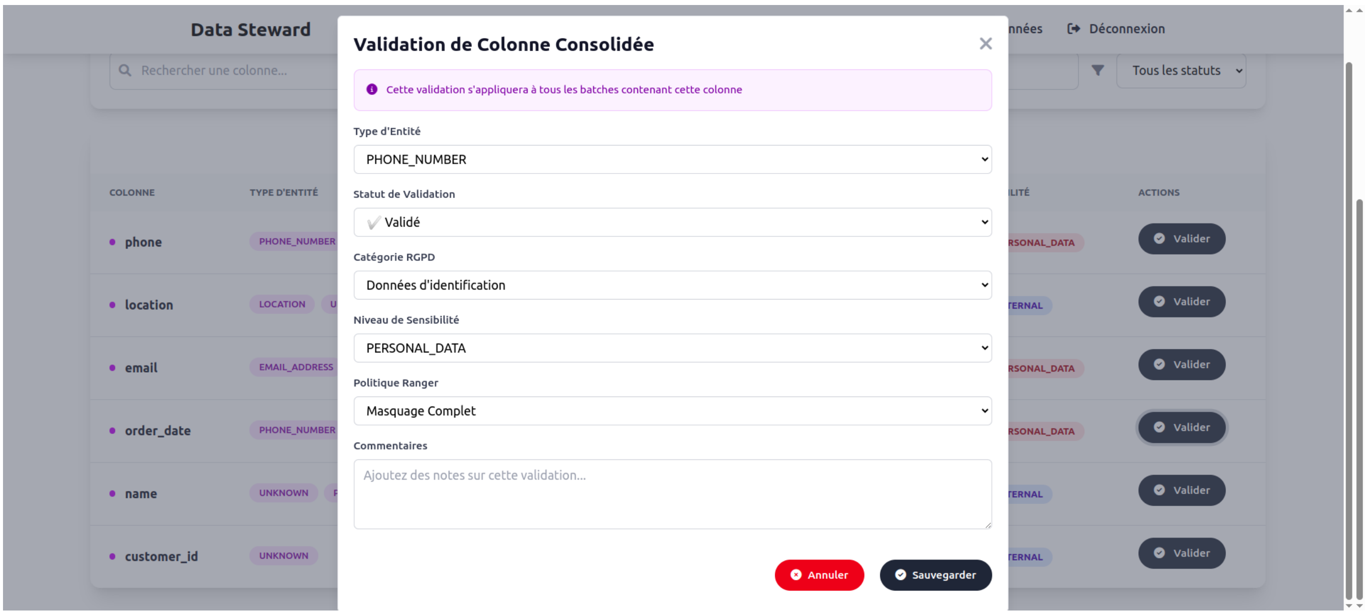

FIGURE 3.34 – Column Validation Modal - Cross-Batch Metadata Approval Workflow

3.4.6 Conclusion

The implementation successfully delivers a comprehensive platform for automatic sensitive data detection and governance through well-designed, role-specific interfaces. The system balances automation with human oversight, ensuring that AI-generated insights enhance rather than replace human decision-making in critical data governance processes.

The modular interface design enables independent evolution of different system components while maintaining consistent user experience and security standards. The integration of AI capabilities (Microsoft Presidio, spaCy NLP, Google Gemini) with traditional data governance workflows creates a powerful platform for enterprise-scale data protection and compliance management.

General Conclusion

This project addressed one of the key challenges of modern enterprises : the automated detection and governance of sensitive personal data in compliance with GDPR regulations. Through the design of an intelligent AI-powered platform, we demonstrated that artificial intelligence can reduce manual work in data protection while improving the accuracy and consistency of compliance workflows.

Summary of Achievements

The developed platform integrates several advanced technologies into a unified ecosystem for data governance. It combines **Microsoft Presidio** for entity recognition—extended with Moroccan-specific recognizers—with **spaCy** for semantic analysis and **Google Gemini** for AI-driven recommendations on compliance, security, data quality, and governance. These results are refined through a **human-in-the-loop** validation workflow, ensuring collaboration between AI and human expertise. Integration with **Apache Atlas** and **Hive** automates glossary generation and metadata management, while **Kafka** enables near–real-time monitoring of sensitive data in enterprise systems.

Contribution and Insights

This work contributes to the field of data governance by showing how AI, NLP, and metadata management can be combined effectively within a scalable governance framework. It validates the relevance of human–AI collaboration in compliance processes, introduces automated glossary generation from validated metadata, and highlights the role of semantic analysis in reducing false detections. On the technical side, customization of Presidio recognizers, optimized batch processing, and MongoDB's flexible schema have proven essential for achieving both scalability and adaptability.

Limitations and Perspectives

Some limitations remain, notably the dependence on French language models, restricted data format support (CSV and JSON), and reliance on external AI services. Future work could include multilingual support, integration of **Apache Ranger** for automated policy enforcement, implementation of **data lineage tracking**, predictive governance using machine learning, and **blockchain-based** audit trails for enhanced transparency.

Broader Reflections

Beyond technical results, this project reflects a paradigm shift where AI transforms compliance from a constraint into an opportunity for trust, efficiency, and quality. Effective governance relies on balance—between automation and human judgment, accuracy and performance, innovation and ethics. As privacy regulations evolve worldwide, the modular and adaptable architecture of this platform positions it to meet these challenges dynamically.

Final Thoughts

This work represents a modest yet meaningful contribution toward intelligent, proactive, and responsible data governance. It illustrates that technology, when thoughtfully designed, can empower humans to protect privacy more effectively. Ultimately, the protection of personal data is not merely a legal obligation—it is a moral imperative and a foundation of digital trust in our modern world.

The protection of personal data is not merely a legal obligation—it is a moral imperative and a foundation of trust in our increasingly digital world.